Manual Therapy 15 (2010) 315–333

Contents lists available at ScienceDirect

Manual Therapy

journal homepage: www.elsevier.com/math

Systematic review

Manipulation or mobilisation for neck pain: A Cochrane Review☆

Anita Gross [a,*], Jordan Miller [a], Jonathan D'Sylva [a], Stephen J. Burnie [b], Charles H. Goldsmith [c], Nadine Graham [a], Ted Haines [c], Gert Brønfort [d], Jan L. Hoving [e], COG [1]

[a] School of Rehabilitation Science, McMaster University, 1400 Main Street West, Hamilton, Ontario L8S 1C7, Canada
[b] Department of Clinical Education, Canadian Memorial Chiropractic College, Toronto, Canada
[c] Department of Clinical Epidemiology & Biostatistics, McMaster University, Hamilton, Canada
[d] Wolfe-Harris Center for Clinical Studies, Northwestern Health Sciences University, Bloomington, MN, USA
[e] Coronel Institute of Occupational Health, Academic Medical Center, Universiteit van Amsterdam, Amsterdam, Netherlands

ARTICLE INFO

Article history:
Received 16 February 2010
Received in revised form
13 April 2010
Accepted 20 April 2010

Keywords:
Systematic review
Neck pain
Manipulation
Mobilisation

ABSTRACT

Manipulation and mobilisation are often used, either alone or combined with other treatment approaches, to treat neck pain. This review assesses if manipulation or mobilisation improves pain, function/disability, patient satisfaction, quality of life (QoL), and global perceived effect (GPE) in adults experiencing neck pain with or without cervicogenic headache or radicular findings. A computerised search was performed in July 2009. Randomised trials investigating manipulation or mobilisation for neck pain were included. Two or more authors independently selected studies, abstracted data, and assessed methodological quality. Pooled relative risk (pRR) and standardised mean differences (pSMD) were calculated. 33% of 27 trials had a low risk of bias. Moderate quality evidence showed cervical manipulation and mobilisation produced similar effects on pain, function and patient satisfaction at intermediate-term follow-up. Low quality evidence suggested cervical manipulation may provide greater short-term pain relief than a control (pSMD −0.90 (95%CI: −1.78 to −0.02)). Low quality evidence also supported thoracic manipulation for pain reduction (NNT 7; 46.6% treatment advantage) and increased function (NNT 5; 40.6% treatment advantage) in acute pain and immediate pain reduction in chronic neck pain (NNT 5; 29% treatment advantage). Optimal technique and dose need to be determined.

© 2010 Elsevier Ltd. All rights reserved.

1. Background

Neck disorders are common, disabling to various degrees, and costly (Côté et al., 1998; Linton et al., 1998; Borghouts et al., 1999; Hogg-Johnson et al., 2008). A significant proportion of direct healthcare costs associated with neck disorders is attributable to visits to healthcare providers, sick leave, and the related loss of productive capacity (Borghouts et al., 1998; Linton et al., 1998; Skargren and Oberg, 1998; Côté et al., 2008). Manipulation and mobilisation are commonly used treatments for neck pain. They have demonstrated both mechanical effects including permanent or short-term change in length of connective tissue and neurophysiologic effects including analgesic effects, motor effects, and sympathetic nervous system effect (Souvlis et al., 2004). Based on our earlier Cochrane Reviews (Gross et al., 1996, 2004a,b) and other reviews (Bodgduk, 2001; Hoving et al., 2001; Peeters et al., 2001; Gross et al., 2002a,b, 2007; Vernon et al., 2006, 2007), studies of their effectiveness have generally been noted when combined with other treatment like exercise and had short-term with inconclusive long-term results. Multimodal approaches that included manual therapy are reported separately (Miller et al., 2010; D'Sylva et al., in press); this review explores manipulation or mobilisation as a single-modal treatment; it has excluded combined therapies. We present a précis of our Cochrane Review's findings using grade summary statements; access Gross et al. (2010) for a detailed report.

2. Objectives

This update of our systematic review assessed the effect of manipulation or mobilisation alone on pain, function, disability, patient satisfaction, global perceived effect (GPE), and quality of life

☆ This paper is based on a Cochrane Review published in The Cochrane Library 2010, Issue 1 (see www.thecochranelibrary.com for information). Cochrane Reviews are regularly updated as new evidence emerges and in response to feedback, and The Cochrane Library should be consulted for the most recent version of the review.

* Corresponding author.
E-mail address: grossa@mcmaster.ca (A. Gross).

[1] COG – Cervical Overview Group: Bronfort G, Burnie SJ, Cameron ID, D'Sylva J, Dumoulin-White R, Eddy A, Ezzo J, Goldsmith CH, Graham N, Gross A, Haines T, Haraldsson B, Hildebrand C, Hoving J, Kay T, Kroeling P, Lilge L, Miller J, Morien A, Peloso P, Perry L, Radylovick Z, Santaguida P, Trinh K, Voth S, Wang E.

1356-689X/$ – see front matter © 2010 Elsevier Ltd. All rights reserved.
doi:10.1016/j.math.2010.04.002

(QoL) in adults experiencing neck pain with or without radicular symptoms and cervicogenic headache.

3. Methods

3.1. Study selection

3.1.1. Types of studies

Any published or unpublished randomised controlled trial (RCT) or quasi-RCT (QRCT), either in full text or abstract form, was included.

3.1.2. Types of participants

The participants were adults (18 years or older) with the following:

- Neck pain without radicular findings, including neck pain without specific cause, whiplash-associated disorder (WAD) categories I and II (Spitzer et al., 1987, 1995; Guzman et al., 2008), myofascial pain syndrome, and neck pain associated with degenerative changes (Schumacher et al., 1993);
- Cervicogenic headache (Headache Classification Committee, 2004; Olesen, 1988; Sjaastad et al., 1990; Olesen and Gobel, 1997); and
- Neck disorders with radicular findings (Rubinstein et al., 2007), including degenerative joint or disc disease with spinal stenosis, spondylolisthesis, or discogenic radiculopathy; and WAD category III (Spitzer et al., 1987, 1995; Guzman et al, 2008).

We defined symptom duration as acute (less than 30 days), subacute (30 days–90 days) or chronic (greater than 90 days).

Studies were excluded if they investigated neck disorders with the following specific causes: a) definite or possible long tract signs (e.g., myelopathies); b) neck pain caused by other pathological entities (Schumacher et al., 1993); c) headache not of cervical origin but associated with the neck; d) co-existing headache when either neck pain was not dominant or the headache was not provoked by neck movements or sustained neck postures; or e) 'mixed' headache, which includes more than one headache classification.

3.1.3. Types of interventions and comparisons

We included studies using either manipulation or mobilisation techniques. Although typically applied to the cervical region, they could also be applied to other body regions; the guiding principle was a mobilisation or manipulation intervention with the intention-to-treat neck pain. Manipulation is a localised force of high velocity and low amplitude directed at specific spinal segments. Mobilisations use low-grade/velocity, small or large amplitude passive movement techniques or neuromuscular techniques within the patient's range of motion and within the patient's control. All studies were compared to either a control or another treatment as follows: a) placebo; b) adjunct treatment, for example: mobilisation plus a treatment (ultrasound) versus that same treatment (ultrasound); c) wait list or no treatment; d) manipulation or mobilisation versus another intervention; e) one technique versus another; or f) one dose versus another dose.

3.1.4. Types of outcome measures

The outcomes of interest were pain relief, function, disability, patient satisfaction, GPE and QoL. We did not set any restriction on the type of tool used in the studies to measure these outcomes as there are no universally accepted tools available; albeit, we found a number of studies did use validated tools. Function and disability could be measured using either self-report measures or observer-based physical performance tests (Beattie, 2001; Finch et al., 2002). Measures of physical performance require testing the subject's ability to execute a simple activity in a standardised environment using a standardised test and scoring procedure; they are concerned with the testing of a coordinated set of functions, which forms a component of functional purposeful activity (i.e. reaching, walking, or driving). Although moderate correlation between self-report scales and physical performance tests exists in the low back literature, it remains unclear if one is superior (Lee et al., 2001) and this relationship in the neck literature remains unclear. We excluded tests used during a standard physical examination such as inspection, range of motion, strength, palpation, provocation, muscular stability, neurological tests, and cervical proprioception. We also extracted data on adverse effects and cost of treatments. The duration of follow-up is defined as: a) immediately post-treatment (within one day); b) short-term follow-up (closest to four weeks); c) intermediate-term follow-up (closest to six months); and d) long-term follow-up (closest to12 months).

3.2. Search methods for identification of studies

A research librarian searched bibliographic databases, without language restrictions, for medical, chiropractic, and allied health literature. We searched CENTRAL; MEDLINE, EMBASE, Manual Alternative and Natural Therapy, Cumulative Index to Nursing and Allied Health Literature (CINAHL), and Index to Chiropractic Literature (ICL) were updated in July 2009. All databases were originally searched from their beginning. We also screened references, personally communicated with identified content experts, and checked our own personal files to identify potential references. Subject headings (MeSH) and keywords included anatomical terms, disorder or syndrome terms, treatment terms, and methodological terms consistent with those advised by the Cochrane Back Review Group. See Appendix 1 for the search strategy for MEDLINE.

3.3. Data collection and analysis

3.3.1. Study selection and data extraction

At least two review authors with expertise in medicine, physiotherapy, chiropractic, massage therapy, statistics, or clinical epidemiology independently conducted citation identification, study selection, and data extraction. Agreement for study selection was assessed using the quadratic weighted Kappa statistic (Kw) with Cicchetti weights (Cicchetti, 1976). A third review author was consulted in the case of persisting disagreement. Pre-piloted forms were used for all phases.

3.3.2. Risk of bias assessment

The Cervical Overview Group uses a calibrated team of interdisciplinary assessors. At least two review authors independently assessed the risk of bias (Appendix 2). The consensus team met to reach a final decision. The following characteristics for risk of bias (maximum criteria = 12, low risk of bias = more than six criteria met) were assessed: randomisation; concealment of treatment allocation; blinding of patient, provider, and outcome assessor; incomplete data: withdrawal/drop-out rate and intention-to-treat analysis; selective outcome reporting; and other: similar at baseline, similar co-interventions, acceptable compliance, similar timing of assessment. Studies were not excluded from further analyses based on the results of risk of bias assessments. We noted explicit details on study design; number analysed and randomised; intention-to-treat analysis; and power analysis in Table 1.

3.3.3. Data analysis

We used descriptive statistics to provide a summarized description of the groups, interventions, outcomes, adverse effect

Table 1
Characteristics of the included studies.

Study/participants	Interventions	Outcomes
Bitterli et al. (1977) Chronic neck disorder with headache or degenerative changes (spondylogenic) n(A/R) 24/30	**Index treatment** Group A (A): technique: manipulation and mobilisation as described by Maigne; frequency: manipulation mean 3–4 session over 3 weeks; dose: 6.2 manipulations over 3.2 sessions and 3.2 sessions of mobilisation; route: cervical spine Group B (B): technique: manipulation as described by Maigne; frequency: 3–4 sessions over 3 weeks; dose: mean 7.2 manipulations over 3.8 sessions; route: cervical spine **Comparison treatment** Group C (C): wait list control **Co-intervention**: none **Duration of treatment**: 3 weeks, 3–4 sessions **Duration of follow-up**: 12 weeks	**Pain** (VAS, 0–100) Baseline mean: A 60.50, B 64.40, C 57.60 End of study mean: A 38.50, B 27.80, C 43.50 Absolute benefit: A 22.0, B 36.6, C 14.1 Reported results: not significant SMD (A v C): −0.18 (95%CI: −1.12 to 0.75) [power 10%] SMD (A v B): 0.36 (95%CI: −0.58 to 1.30) [power 7%] SMD (B v C): −0.54 (95%CI: −1.43 to 0.36) [power 10%] **Function**: NR **GPE**: NR **Patient satisfaction**: NR **QoL**: NR **Side effects**: manipulation and mobilisation were well tolerated with the customary reaction of minimal benign reaction lasting less than 24 h; RR: 1.34 (95%CI: 0.77–2.34) **Cost of care**: NR
Cassidy et al. (1992) Acute, subacute, chronic neck pain without radicular findings and with varied spondylosis n(A/R) 100/100	**Index treatment** Manipulation (manip): technique: rotation manipulation away from the direction of pain; frequency: one session; dose: one manipulation; route: cervical spine **Comparison treatment** Mobilisation (mob): technique: muscle energy technique described by Bourdillon, an isometric contraction localised to the involved level; frequency: one session; dose: 4 repetitions held for 5 s; route: cervical spine **Co-intervention**: none **Duration of treatment**: one session **Duration of follow-up**: none	**Pain** (NRS101, 0–100) Baseline mean: manip 37.7, mob 31.0 End of study mean: manip 20.4, mob 20.5 Absolute benefit: manip 17.3, mob 10.5 Reported results: not significant SMD: 0.00 (95%CI: −0.40 to 0.39) [power 16%] **Function**: NR **Patient satisfaction**: NR **GPE**: NR **QoL**: NR **Side effects**: no complications; 3 subjects in each group reported more pain after intervention; RR: 0.97 (95%CI: 0.47–2.03) **Cost of care**: NR
Chen et al. (2007) Chronic cervicogenic headache from degenerative changes n(A/R) 65/70	**Index treatment** Manipulation (manip): technique: manipulation; frequency: 10 session, every other day; dose: 20–30 min treatment, route: cervical spine **Comparison treatment** Transcutaneous Electrical Nerve Stimulation (TENS): technique: TENS Perimedic, 10Hz, 250ns; frequency: 10 session every other day; dose: 20 min duration; route: NR **Co-intervention**: NR **Duration of treatment**: 10 sessions **Duration of follow-up**: 1 week	**Pain** (NRS, 0–10) Baseline mean: manip 7.45, TENS 7.86 End of study mean: manip 2.31, TENS 5.26 Absolute benefit: manip 5.41, TENS 2.6 Reported results: significant favouring manipulation SMD: −1.91 (95%CI: −2.50 to −1.31) **Function**: NR **Patient satisfaction**: NR **GPE**: NR **QoL**: NR **Side effects**: none **Cost of care**: NR
Cleland et al. (2005) Chronic neck pain n(A/R) 36/36	**Index treatment** Manipulation (manip): technique: thoracic manipulation to each restricted segment; frequency: one session; dose: 1–2 manipulations per segment; average 3.7 manipulations per subject; route: thoracic spine **Comparison treatment** Placebo Manipulation (cntl): technique: use of flat open hand on subject naive to manipulation; frequency: one session; dose: average 3 placebo manipulations per subject **Co-intervention**: NR **Duration of treatment**: one session **Duration of follow-up**: none	**Pain** (VAS, 0–100) Baseline mean: manip 41.6, cntl 47.7 End of study mean: manip 26.1, cntl 43.5 Absolute benefit: manip 15.5, cntl 4.2 Reported results: significant favouring manipulation SMD: −0.93 (95%CI: −1.62 to −0.24) **Function**: NR **Patient satisfaction**: NR **GPE**: NR **QoL**: NR **Side effects**: none **Cost of care**: NR
Coppieters et al. (2003) Subacute, chronic mechanical neck disorder without radicular findings n(A/R) 20/20	**Index treatment** Experimental group: technique: mobilisations including lateral glide techniques described by Elvey, the arm is progressively positioned from an unloaded to a preloaded position; frequency: one session; dose: mean 4.5 min treatment duration, session one grade two technique, session 2–3, grade 3 technique; route: C5 and C6 most frequently treated **Comparison treatment** Control Group (cntl): pulsed ultrasound **Co-intervention**: NR **Duration of treatment**: one session **Duration of follow-up**: none	**Pain** (NRS101, 0–100) Baseline mean: experimental 7.3, cntl 7.7 End of study mean: experimental 5.8, cntl 7.4 Absolute benefit: experimental 1.5, cntl 0.3 Reported results: positive immediate effect favouring mobilisations SMD: −0.78 (95%CI: −1.70 to 0.13) [power 58%] **Function**: NR **Patient satisfaction**: NR **GPE**: NR **QoL**: NR **Side effects**: NR **Cost of care**: NR
David et al. (1998) Subacute, chronic neck pain with	**Index treatment** Physiotherapy group (PT): technique: passive mobilisation:	**Pain** (VAS, 0–100) Baseline mean: PT 51, A 51

(*continued on next page*)

Table 1 (*continued*)

Study/participants	Interventions	Outcomes
degenerative changes [cervical spondylosis], whiplash-associated disorder n(A/R) 51/70	standard localised mobilisation described by Maitland, rotation, posterior–anterior oscillations, longitudinal traction; frequency: one session/week; dose: NR; route: cervical spine **Comparison treatment**: Acupuncture Group (Acup): technique: local needling of trigger point, regional needling (GB21-supraspinatus tender area), distal needling (LI4-web space between thumb and first finger); frequency: one session/week; dose: needle left in situ for 15 min **Co-intervention**: NR **Duration of treatment**: 6 weeks, maximum 6 sessions **Duration of follow-up**: 24 weeks	End of study mean: PT 22, acup 28 Absolute benefit: PT 29, acup 23 Reported results: not significant SMD: −0.33 (95%CI: −0.88 to 0.23) [power 24%] **Function** (Northwick Pain Questionnaire, 0–36) Baseline mean: PT 36, acup 36 End of study mean: PT 22, acup 25 Absolute benefit: PT 14, acup 11 Reported results: not significant SMD: −0.16 (95%CI: −0.72 to 0.39) [power 15%] **Patient satisfaction**: NR **GPE**: NR **QoL** (GHQ 28, 0–28): Reported results: not significant RR1.07 (0.48–2.35) **Side effects**: no side effects occurred for acupuncture **Cost of care**: NR
Egwu (2008) Acute or subacute neck pain − neck disorder associated with degenerative changes − cervical spondylosis of C5/6 n(A/R) 95/96	**Index treatment groups** Posterior–anterior–unilateral pressure (PAUP): technique: posterior–anterior oscillatory pressure on the side of symptoms; frequency: 3 times/week; dose: 10 oscillations in 60 s; route: transverse processes of C5/6 Anterior–posterior–unilateral pressure (APUP): technique: anterior–posterior oscillatory pressure on the side of symptoms; frequency: 3 times/week; dose: 10 oscillations in 60 s; route: transverse processes of C5/6 Cervical oscillatory rotation (COR): technique: rotation oscillation by turning head and neck to the opposite side from the pain until pain is just elicited and then oscillations are given; frequency: 3 times/week; dose: 10 oscillations in 60 s; route: rotation performed with hands on jaw and occiput Transverse oscillatory pressure (TOP): technique: transverse oscillations produced by a push relax sequence on the spinous process using the thumbs to produce the movement; frequency: 3 times/week; dose: 10 oscillations in 60 s; route: spinous processes of C5/6 **Co-intervention**: NR **Duration of therapy period**: 4 weeks or until pain was absent, 12 sessions maximum **Duration of follow-up**: No follow-up	**Pain status** (5 categories, pain free to worse) Reported results: NR RR (APUP v PAUP): 0.60 (95%CI: 0.16–2.23) RR (APUP v COR): 0.29 (95%CI: 0.09–0.91) RR (APUP v TOP): 0.30 (95%CI: 0.09–0.96) RR (PAUP v COR): 0.48 (95%CI: 0.19–1.19) RR (PAUP v TOP): 0.50 (95%CI: 0.20–1.25) **Mean treatment time** (min) Mean: PAUP 161, APUP 130, COR 201, TOP 221 Reported results: not significant Relapse after 3 months (Number reported) PAUP 0, APUP 0, COR 3, TOP 2 Reported results: NR **Function**: NR **GPE**: NR **Patient satisfaction**: NR **QoL**: NR **Side effects**: NR **Cost of care**: NR
Fernandez-de-las-Penas et al. (2004) Acute/subacute WAD II and III n(A/R) 88/88	**Index treatment** Group A: technique: thoracic manipulation and same treatment as group B; frequency: 15 sessions over 3 weeks; dose: 2 manipulations over 2 weeks; route: thoracic spine **Comparison treatment** Group B: technique: ultrasound to neck, active exercises at home, muscle stretching, massage, multimodal care, electrotherapy, manual therapies; frequency: 15 sessions over 3 weeks **Co-intervention**: NR **Duration of treatment**: 3 weeks, 15 sessions **Duration of follow-up**: none	**Pain** (VAS, 0–10) Baseline mean: NR End of study mean: NR Absolute benefit: A 2.27, B 1.66 Reported results: significant favouring group A SMD: −0.68 (95%CI: −1.11 to −0.25) NNT: unable to calculate due to baseline data not reported **Function**: NR **Patient satisfaction**: NR **GPE**: NR **QoL**: NR **Side effects**: NR **Cost of care**: NR
Giles & Muller (1999) Chronic neck pain with degenerative changes n(A/R) 98/157 for all spinal patients; 62/? for neck subgroup	**Index treatment** Manipulation (manip): technique: high velocity, low amplitude manipulation; frequency: 6 sessions/median 19 days; dose: 15–20 min appointments; route: cervical spine **Comparison treatment** Acupuncture (acup): technique: needles with low-volt electrical stimulation to tender points; frequency: 6 sessions/median 40 days; dose: 8–10 needles Medication (med): tenoxicam (NSAID) with ranitidine, median 15 days **Co-intervention**: NR **Duration of treatment**: 3–4 weeks, 6 sessions **Duration of follow-up**: none	**Pain** (change scores, VAS, 0 to10) Baseline median: manip 4.5, acup 2.0, med 4.0 Absolute benefit: manip 1.5, acup 1.0, med 0.5 Reported results: not clear SMD (manip v med): −0.35 (95%CI: −1.05 to 0.35) [power 5%] SMD (manip v acup): −0.23 (95%CI: −1.04 to 0.50) [power 7%] **Function** (NDI, 0–50) Baseline median: manip 32, acup 40, med 28 Absolute benefit: manip 10.0, acup 6.0, med 0.0 Reported results: not clear SMD (manip v med): −0.92 (95%CI: −1.74 to −0.10) SMD (manip v acup): −0.33 (95%CI: −1.12 to 0.46) [power 7%] **Patient satisfaction**: NR **Side effects**: no side effects occurred for acup or manip **Cost of care**: NR

Table 1 (*continued*)

Study/participants	Interventions	Outcomes
Gonzalez-Iglesias et al. (2009) Acute neck pain n(A/R) 45/45	**Index treatment** Thoracic spine manipulation (TSM): technique: seated distraction manipulation – subject has arms across chest, therapist gently flexes the thoracic spine until tension is felt and then a distraction thrust manipulation is applied. If no pop was heard, a second attempt was made; frequency: 1 session/week; dose: grade 5 manipulation; route: thoracic spine Electrothermal therapy (ETT): type: infrared lamp and TENS; frequency: 2 sessions/week; dose: Infrared 250W for 15 min, TENS 100Hz for 20 min; route: infrared lamp 50 cm from the subject's neck, TENS applied to each side of C7 **Comparison treatment** Electrothermal therapy (ETT): Type: infrared lamp and TENS; frequency: 2 sessions/week; dose: Infrared 250W for 15 min, TENS 100Hz for 20 min; route: infrared lamp 50 cm from the subject's neck, TENS applied to each side of C7 **Co-intervention**: NR **Duration of treatment**: 3 weeks, 5 sessions **Duration of follow-up**: 4 weeks	**Pain** (at rest change score, VAS, 0–100) Baseline mean: TSM + ETT 54.7, ETT 52.7 Final treatment mean: TSM + ETT 20.2, ETT 44.7 2 week follow-up mean: TSM + ETT 26.4, ETT 41.2 4 week follow-up mean: TSM + ETT 21.5 ETT 42.2 Reported results: significant for comparisons at all time points Final treatment SMD (TSM + ETT v ETT): −3.43 (95%CI: −4.38 to −2.49) 4 week follow-up SMD (TSM + ETT v ETT): −2.19 (95%CI: −2.94 to −1.44) **Function** (Northwich Pain Questionnaire, 0–36) Baseline mean: TSM + ETT 27.9, ETT 27.0 Final treatment mean: TSM + ETT 15.2, ETT 23.1 2 week follow-up mean: TSM + ETT 14.7, ETT 21.8 Reported results: significant for comparisons at all time points Final treatment SMD (TSM + ETT v ETT): −2.17 (95%CI: −2.92 to −1.42) 2 week follow-up SMD (TSM + ETT v ETT): −2.28 (95%CI: −3.05 to −1.52) **GPE**: NR **Patient satisfaction**: NR **QoL**: NR **Side effects**: NR **Cost of care**: NR
Haas et al. (2004) Chronic neck pain with headache n(A/R) 25/26	**Index treatment** Group 3s/w (9s): technique: manipulation: high velocity, low amplitude manipulation plus heat, soft-tissue therapy with massage, trigger point therapy, education (modification of ADL), rehabilitation exercises; frequency: 3 sessions per week; dose: NR; route: cervical spine Group 4s/w (12s): technique: manipulation: high velocity, low amplitude manipulation plus heat, soft-tissue therapy with massage, trigger point therapy, education (modification of ADL), rehabilitation exercises; frequency: 4 sessions per week; dose: NR; route: cervical spine **Comparison treatment** Group 1s/w (3s): technique: manipulation: high velocity, low amplitude manipulation plus heat, soft-tissue therapy with massage, trigger point therapy, education (modification of ADL), rehabilitation exercises; frequency: 1 session per week; dose: NR; route: cervical spine **Co-intervention**: heat, soft-tissue therapy with massage, trigger point therapy, education (modification of ADL), rehabilitation exercises **Duration of treatment**: 3 weeks, 9–12 sessions **Duration of follow-up**: 1 week post-treatment, 11 weeks post-treatment	**Pain** (modified von Korff scale, average 3 scales, 0–100) Baseline mean: 12s 49.6, 9s 58.7, 3s 61.0 Absolute benefit: 12s 18.8, 9s 31.7, 3s 18.6 Reported results: findings give preliminary support for larger doses 9–12 sessions SMD (12s v 3s): −0.48 (95%CI: −1.51 to 0.56) [power 64%] SMD (12s v 9s): 0.17 (95%CI: −0.81 to 1.16) [power 60%] SMD (9s v 3s): −0.90 (95%CI: −1.98 to 0.18) [power 69%] **Function** (neck disability modified von Korff scale, average 3 scales, 0–100) Baseline mean: 12s 33.8, 9s 35.7, 3s 46.7 Absolute benefit: 12s 20.1, 9s 22.5, 3s 13.4 Reported results: findings give preliminary support for larger doses 9–12 sessions SMD (12s v 3s): −1.15 (95%CI: −2.27 to −0.03) SMD (12s v 9s): −0.03 (95%CI: −1.01 to 0.95) [power 056] SMD (9s v 3s): −1.47 (95%CI: −2.65 to −0.28) **Patient satisfaction**: NR **GPE**: NR **QoL**: NR **Side effects**: none **Cost of care**: NR
Howe et al. (1983) Subacute, chronic neck pain with radicular findings and headache n(A/R) 44/52	**Index treatment** Manipulation (manip) Group: technique: manipulation and same treatment as in control group; frequency: 1 session; dose: up to 3 manipulation; **Comparison treatments** Control group (med): azapropazone; dose unknown **Co-intervention**: 2 subjects had lignocaine-hydrocortisone injections in manipulation group **Duration of treatment**: 1 session **Duration of follow-up**: 3 weeks	**Pain** (count, neck/shoulder/arm/hand pain and headache present) Baseline mean: NR Reported results: significant favouring manipulation RR: 0.56 (95%CI: 0.19–1.68) [power 18%] **Function**: NR **Patient satisfaction**: NR **Side effects**: NR **Cost of care**: NR
Hurwitz et al. (2002) Subacute and chronic neck pain with or without radicular findings and headache n(A/R) 269/336	**Index treatments** Manipulation (manip): technique: manipulation-dynamic thrust applied with high velocity low amplitude force with minimal extension and rotation, directed at 1 or more restricted upper thoracic or cervical spine joint segments; frequency: NR; dose: at least one manipulation; route: cervical spine Manipulation with heat (manip/heat): 10 min moist heat application before manipulation; Manipulation with electrical muscle stimulation (manip/EMS): 10 min application of this modality before manipulation; parameters NR Mobilisation (mob): 1 or more low velocity, variable	**Pain** (NRS, 0 to10) Baseline mean: NR for each subgroup End of study mean: NR for each subgroup Reported results: no significant difference SMD (manip v mob): 0.15 (95%CI: −0.32 to 0.61) SMD (manip + heat + EMS v mob): −0.28 (95%CI: −0.77 to 0.21) SMD (mod + heat + EMS v manip): 0.24 (95%CI: −0.24 to 0.71) RR (heat v no heat): 1.14 (95%CI Mixed: 0.95–1.37) RR (EMS v no EMS): 0.90 (95%CI Mixed: 0.73–1.13) **Function** (NDI, 0–50) Baseline mean: NR for each subgroup

(*continued on next page*)

Table 1 (*continued*)

Study/participants	Interventions	Outcomes
	amplitude movements applied within the patient's passive range of motion directed to 1 or more restricted upper thoracic or cervical spine joint segments Mobilisation with heat (mob/heat): 10 min moist heat application before mobilisation Mobilisation with EMS (mob/EMS): 10 min application of this modality before mobilisation; parameters NR **Comparison treatments** Any of the above noted treatment combinations **Co-intervention**: All participants received information on posture and body mechanics and one or more of the following: stretching, flexibility, or strengthening exercises and advice about ergonomic and workplace modifications **Duration of therapy period**: NR (approximately 6 weeks) **Duration of follow-up**: 6 months	End of study mean: NR for each subgroup Reported results: no significant difference SMD (manip v mob): 0.07 (95%CI: −0.40 to 0.54) SMD (manip + heat + EMS v mob): −0.08 (95%CI: −0.56 to 0.41) SMD (mod + heat + EMS v manip): 0.14 (95%CI: −0.33 to 0.62) RR (heat v no heat) 1.14 (95%CI Mixed: 0.94–1.38) RR (EMS v no EMS) 0.87 (95%CI Mixed: 0.69–1.10) **Patient satisfaction** (10–50 scale; at 4w of care) Reported results: no significant difference SMD (manip v mob): 0.11 (95%CI: −0.35 to 0.58) SMD (manip + heat + EMS v mob): 0.14 (95%CI: −0.35 to 0.62) SMD (mod + heat + EMS v manip): −0.12 (95%CI: −0.59 to 0.36) **Side effects**: interviewed at 4 weeks of care, no known study related adverse events; manipulation group had statistically significant more transient minor discomfort (16%) v mobilisation group (8.7%) **Cost of care**: number of disability days were not significantly different between groups
Kanlayanaphotporn et al. (2009) Chronic neck pain n(A/R) 60/60	**Index treatment** Preferred mobilisation (PMob): technique: posterior–anterior pressure on the side of symptoms; frequency: 1 session; dose: 2× 1 min mobilisations graded 1–2 for pain and 3–4 for stiffness; route: articular processes of cervical spine **Comparison treatment** Random mobilisation (RMob): technique: 1 of 3 techniques selected i) central posterior–anterior pressure ii) ipsilateral posterior–anterior pressure iii) contralateral posterior–anterior pressure; frequency: 1 session; dose 2× 1 min mobilisations; route: articular processes of cervical spine **Co-intervention**: NR **Duration of therapy period**: 1 session **Duration of follow-up**: No follow-up	**Pain** (at rest change score, VAS, 0–100) Baseline mean: PMob 47.4, RMob 48.3 Absolute benefit: PMob 10.8 RMob 12.3 Reported results: not significant SMD (PMob v RMob): −0.01 (95%CI: −0.52 to 0.49) **Pain** (with most painful movement change score, VAS, 0–100) Baseline mean: PMob 59.5, RMob 61.6 Absolute benefit: PMob 16.7 RMob 16.9 Reported results: not significant SMD (PMob v RMob): −0.12 (95%CI: −0.63 to 0.38) **Function**: NR **GPE** (1–7): Reported results: not significant RR (PMob v RMob): 1.12 (95%CI: 0.74–1.69) **Patient satisfaction**: NR **QoL**: NR **Side effect**: NR **Cost of care**: NR
Kanlayanaphotporn et al. (2010) Chronic neck pain n(A/R) 60/60	**Index treatment** Central Posterior–Anterior Mobilisation (PAMob): technique: posterior–anterior pressure over the spinous process of restricted segments; frequency: 1 session; dose: 2× 1 min mobilisations graded 1–2 for pain and 3–4 for stiffness; route: spinous processes of cervical spine **Comparison treatment** Random mobilisation (RMob): technique: 1 of 3 techniques selected i) central posterior–anterior pressure ii) ipsilateral posterior–anterior pressure iii) contralateral posterior–anterior pressure; frequency: 1 session; dose: 2× 1 min mobilisations; route: articular processes of cervical spine **Co-intervention**: NR **Duration of therapy period**: 1 session **Duration of follow-up**: No follow-up	**Pain** (at rest change score, VAS, 0–100) Baseline mean: PAMob 50.2, RMob 47.3 Absolute benefit: PAMob 18.3 RMob 13.0 Reported results: not significant SMD (PAMob v RMob): 0.32 (95%CI: −0.18 to 0.83) **Pain** (with most painful movement change score, VAS, 0–100) Baseline mean: PAMob 60.8, RMob 59.8 Absolute benefit: PAMob 21.9 RMob 12.7 Reported results: not significant SMD (PAMob v RMob): 0.53 (95%CI: 0.01–1.04) **Function**: NR **GPE** (1–7): Reported results: not significant RR (PAMob v RMob): 0.90 (95%CI: 0.63–1.30) **Patient satisfaction**: NR **QoL**: NR **Side effects**: NR **Cost of care**: NR
Krauss et al. (2008) Neck pain (duration of symptoms NR) n(A/R) 32/32	**Index treatment** Thoracic spine manipulation (TSM): technique: bilateral translatoric facet joint traction manipulation to the upper thoracic intervertebral segment performed; frequency: 1 session; dose: grade 5 manipulation; route: thoracic spine **Comparison treatment** No treatment (NT): no treatment provided **Co-intervention**: NR **Duration of therapy period**: 1 session **Duration of follow-up**: no follow-up	**Pain** (intensity with left rotation, 9 point faces pain scale) Baseline mean: TSM 3.73, NT 2.50 Absolute benefit: TSM 0.688 NT 0.667 Reported results: not significant SMD (TSM v NT): 0.02 (95%CI: −0.73 to 0.77) **Pain** (intensity with right rotation, 9 point faces pain scale) Baseline mean: TSM 2.75, NT 2.80 Absolute benefit: TSM 1.5, NT −0.1 Reported results: not significant SMD (TSM v NT): 0.65 (95%CI: −0.12 to 1.41) **Function**: NR **GPE**: NR **Patient satisfaction**: NR

Table 1 (*continued*)

Study/participants	Interventions	Outcomes
		QoL: NR **Side effects**: NR **Cost of care**: NR
Martinez-Segura et al. (2006) Subacute, chronic neck pain n(A/R) 71/71	**Index treatment** Cervical high velocity low amplitude technique group (manip): technique: manipulation ipsilateral side flexion, contralateral rotation; frequency: 1 session; route: cervical spine **Comparison treatment** Sham/Control Group (cntl): technique: neck positioned into ipsilateral side flexion, contralateral rotation, position held for 30 s, the side of manual contact was randomised; frequency: 1 session **Co-intervention**: NR **Duration of therapy period**: 1 session **Duration of follow-up**: none	**Pain** (VAS, 0–10 cm) Baseline mean: manip 5.7, cntl 5.5 End of study mean: manip 2.2, cntl 5.1 Absolute benefit: manip 3.5, cntl 0.4 Reported results: significant favouring manipulation SMD: −1.67 (95%CI: −2.21 to −1.12) **Function**: NR **Patient satisfaction**: NR **GPE**: NR **QoL**: NR **Side effects**: NR **Cost of care**: NR
Muller and Giles et al. (2005) Chronic neck pain n(A/R) 65/116	**Index treatment** Spinal manipulation (SM): technique: high velocity low amplitude thrust on the level of involvement; frequency: 2 sessions/week; dose: grade 5 manipulation, 20 min visits; route: affected segments of cervical spine **Comparison treatment** Medication (med): type: celebrex-celacoxin (27 subjects), vioxx-rofecoxib (11 subjects), paracetamol (5 subjects); frequency: fortnightly 20 min office visits until asymptomatic or sufficient pain relief achieved; dose celebrex-celacoxin 200–400 mg/day, vioxx-rofecoxib 12.5–25 mg/day, paracetamol 1000–3000 mg/day; route: oral. Acupuncture (AP): technique: 8–10 needles placed in local paraspinal and intramuscular pain areas. 5 needles placed in distal point meridians; frequency: 2 sessions/week until asymptomatic or acceptable pain relief achieved; dose: 20 min with turning or flicking needles every 5 min if tolerated; route: sterile HWATO Chinese acupuncture guide tube needles (50 mm length, 0.25 mm gauge). **Co-intervention**: not avoided **Duration of therapy period**: 9-weeks or until asymptomatic, maximum 18 sessions **Duration of follow-up**: 12 months	**Pain** (VAS, 0–10) Baseline Median: SM 6, med 4, AP 7 12 months Median: SM 2.8, med 4.7, AP 2.5 Reported results: significance between groups NR, significant for SM and AP within group data SMD (SM v med): −0.24 (95%CI: −0.85 to 0.37) SMD (SM v AP): 0.04 (95%CI: −0.56 to 0.64) **Pain** (frequency, days/week) Baseline Median: SM 4, med 2, AP 4 12 months Median: SM 2, med 4, AP 3 Reported results: significance between groups NR, significant within the SM group only SMD (SM v med): −0.55 (95%CI: −1.17 to 0.07) SMD (SM v AP): −0.22 (95%CI: −0.82 to 0.38) **Function** (NDI, 0–50) Baseline Median: SM 28, med 42, AP 36 12 months Median: SM 20, med 36, AP 24 Reported results: significance between groups NR, significant within the SM and AP groups SMD (SM v med): −0.36 (95%CI: −0.97 to 0.25) SMD (SM v AP): −0.09 (95%CI: −0.69 to 0.51) **GPE**: NR **Patient satisfaction**: NR **QoL**: NR for neck group alone **Side effects**: NR **Cost of care**: NR
Nilsson et al. (1997) Chronic neck pain with headache n(A/R) 53/54	**Index treatment** Manipulation Group (manip): technique: manipulation − toggle recoil for upper cervical spine, diversified technique for mid- and lower cervical spine; high velocity, low amplitude thrust at the end point of passive range of motion; frequency: 2 sessions/week; dose: 12 toggle recoil, 10 diversified manipulation; route: cervical spine **Comparison treatment** Soft-Tissue Group (ST): technique: massage, deep frictions and trigger point treatment of posterior muscles of shoulder girdle, upper thoracic and lower cervical and placebo laser applied to the upper cervical region; frequency: 2 session/week; route: cervical and thoracic spine **Co-intervention**: NR **Duration of therapy period**: 3 weeks, 6 sessions **Duration of follow-up**: 1 week	**Pain** (headache intensity per episode, VAS, 0–100) Baseline Median: manip 48, ST 37 End of Study Median: manip 15, ST 6 Absolute benefit: manip 33, ST 31 Reported results: significant favouring manipulation SMD: −0.45 (95%CI: −0.99 to 0.10) [power 16%] **Function**: NR **Patient satisfaction**: NR **Side effects**: NR **Cost of care**: NR
Parkin-Smith and Penter (1998) Neck pain, duration disorder NR n(A/R) 30/30	**Index treatment**: Group A: technique: manipulation; route: cervical spine **Comparison treatment**: Group B: technique: manipulation; route: cervical and thoracic spine **Co-intervention**: not specified **Duration of therapy period**: 3 weeks, 6 sessions **Duration of follow-up**: none	**Pain** (NRS101, 0–100) Baseline mean: A 33.89, B 33.00 End of study mean: A 17.17, B 13.18 Absolute benefit: A 16.72, B 19.82 Reported results: not significant SMD: 0.29 (95%CI: −0.43 to 1.01) [power 94%] **Function** (Neck Disability Index, 0–50) Baseline mean: A 18.24, B 17.64 End of study mean: A 6.89, B 4.71 Absolute benefit: A 11.35, B 12.93 Reported results: not significant SMD: 0.30 (95%CI: −0.42 to 1.02) [power 100%]

(*continued on next page*)

Table 1 (*continued*)

Study/participants	Interventions	Outcomes
		Patient satisfaction: NR **Side effects**: NR **Cost of care**: NR
Savolainen et al. (2004) Neck pain, duration disorder NR, radicular findings NR n(A/R) 41/75	**Index treatment** Thoracic manipulation (manip): technique: manipulation – upper thoracic spine by a physiatrist; frequency: 1 session/week; dose: four manipulations; route: thoracic spine **Comparison treatment** Instructed exercise (ex): duration unclear **Co-intervention**: NR Duration of Therapy Period: 4 weeks for manipulation group, unknown for control group **Duration of follow-up**: 52 weeks	**Pain** (pain right now; VAS, 0–10) Baseline mean: manip 3.7, ex 3.8 Absolute benefit: manip 0.8, ex −0.5 Results: significant favouring thoracic manipulation for perceived worst pain, no other significant results between groups for pain right now and average pain SMD (pain right now): −0.50 (−1.13, 0.13) [power 69%] **Function**: NR **Patient satisfaction**: NR **GPE**: NR **QoL**: NR **Side effects**: NR **Cost of care**: NR
Sloop et al. (1982) Subacute, chronic neck pain with variable degenerative changes [cervical spondylosis] n(A/R) 39/39	**Index treatment** Manipulation Group (manip): technique: manipulation described by Cyriax, Maigne, Maitland, Matthews, muscle relaxant; frequency: 1 session; route: cervical spine **Comparison treatment**: Control Group (cntl): muscle relaxant **Co-intervention**: "other medical management was not restricted during the study" **Duration of therapy period**: 1 session **Duration of follow-up**: 3 weeks (then crossover occurs)	**Pain** (VAS, 0–100) Baseline: NR Absolute benefit: MT 18, cntl 5 Reported results: not significant SMD: 0.40 (95%CI: −0.23 to 1.04) [power 5%] **Function** (selected daily activities, VAS 0–100) Baseline: NR Reported results: not significant **Global perceived effect** [patient perceived effect, 0 (completely well) to 8 (worst possible) collapsed to dichotomous response (improved/not improved)] Reported results: not significant RR: 0.59 (95%CI: 0.34–1.05) **Side effects**: 2 people had superficial phlebitis following diazepam injection and recovered uneventfully; 2 people in manipulation group reported a new discomfort in their necks followed by improvement in their chronic neck pain; RR: 1.0 (95%CI: 0.4–2.4) **Cost of care**: NR
Strunk and Hondras (2008) Subacute or chronic neck pain n(A/R) 5/6	**Index treatment** Combined therapeutic approach (CTA): thoracic spine and sacroiliac joint manipulation and muscle energy techniques Thoracic spine and sacroiliac joint manipulation: technique: high velocity low amplitude spinal manipulation to thoracic spine and sacroiliac joint; frequency: 2 sessions/week; dose: grade 5 manipulation; route: thoracic spine and sacroiliac joint Muscle energy: technique: post isometric relaxation technique to hypertonic muscles according to Lewis Procedures; frequency: 2 sessions/week; dose: 2 sets of 3 repetitions; route: left or right scalenes, upper fibres of trapezius, levator scapulae, suboccipital muscles based on assessment findings **Comparison treatment** Cervical high velocity, low amplitude manipulation (CHVLA): Technique: high velocity, low amplitude manipulation of the hypomobile segment of the cervical spine; frequency: 2 sessions/week; dose: grade 5 manipulation; route: hypomobile segments identified from C0–C7 **Co-intervention**: NR **Duration of therapy period**: 2 weeks, 4 sessions **Duration of follow-up**: no follow-up	**Pain** (VAS, 0–100) Baseline median: CTA 35, CHVLA 29 2 weeks median: CTA 65, CHVLA 27 Reported results: NR SMD (CTA v CHVLA): −0.99 [95%CI: −2.85, 0.88] **Function** (NDI, 0–100%) Baseline median: CTA 34, CHVLA 24 2 weeks median: CTA 26, CHVLA 20 Reported results: NR SMD (CTA v CHVLA): −0.57 [95%CI: −2.27, 1.12] **GPE**: NR **Patient satisfaction**: NR **QoL**: NR **Side effects**: discomfort: CTA and cervical HVLA combined 3/5 neck pain/stiffness (lasting <10 min): CTA 0/3, Cervical HVLA 1/2 dizziness/imbalance: CTA 0/3, cervical HVLA 1/2 neck pain/stiffness (onset >24 h post-treatment, duration >24 h) CTA 1/3, cervical HVLA 0/2 **Cost of care**: NR
van Schalkwyk and Parkin-Smith (2000) Neck pain, disorder duration NR n(A/R) 30/30	**Index treatment** **Group A**: technique: cervical rotary break manipulation with contact taken on the ipsilateral side, described by Szaraz; frequency: 10 sessions over 4 weeks; route: cervical spine **Comparison treatment** **Group B**: technique: lateral break manipulation with contact taken on the contralateral side, described by Szaraz; frequency: 10 sessions over 4 weeks; route: cervical spine **Co-intervention**: NR **Duration of therapy period**: 4 weeks, 10 sessions **Duration of follow-up**: 4 weeks	**Pain** (NRS101, 0–100) Baseline mean: Group A 38.28, Group B 33.25 End of study mean: Group A 9.40, Group B 17.54 Absolute benefit: Group A 28.88, Group B 15.71 Reported results: not significant; however our calculations show the difference to favour Group A SMD −0.82 (95%CI: −1.57 to −0.07) [power 96%] **Function** (NDI, 0–50) Baseline mean: Group A 22.53, Group B 16.4 End of study mean: Group A 6.00, Group B 6.13 Absolute benefit: Group A 16.53, Group B 10.27 Reported results: not significant [power 34%] SMD −0.01 (95%CI: −0.72 to 0.71) **Side effects**: NR **Cost of care**: NR

Table 1 (*continued*)

Study/participants	Interventions	Outcomes
Vernon et al. (1990) Acute, subacute, chronic neck pain n(A/R) 9/9	**Index treatment** Manipulation (M): technique: manipulation– rotational, high velocity, low amplitude thrust; frequency: 1 session; route: cervical spine **Comparison treatment**: Sham/Mobilisation (S): technique: rotational mobilisation with gentle oscillations into elastic barrier; technique described by Sandoz; frequency: 1 session; route: cervical spine **Co-intervention**: NR **Duration of treatment**: 1 session **Duration of follow-up**: none	**Pain** [pain pressure threshold, algometry (kg/cm)] Baseline mean: M 3.4, S 2.3 End of study mean: M 4.8, S 2.3 Absolute benefit: M 1.4. S 0.0 Reported results: significant favouring manipulation when assessed by ANOVA SMD (point 2): −1.15 (95%CI: −2.65 to 0.34) [power 5%] **Function**: NR **Patient satisfaction**: NR **Side effects**: no complications **Cost of care**: NR
Wood et al. (2001) Subacute mechanical neck pain n(A/R) 30/30	**Index treatment** Group A: technique: manipulation using Activator II Adjusting Instrument, mechanical force, manually assisted; frequency: 2–3 sessions/week; route: cervical spine **Comparison**: Group B: technique: manual manipulation, high velocity, low amplitude manual adjustment; frequency: 2–3 sessions/week; route: cervical spine **Co-intervention**: no medication for at least one month, no other treatment modalities, exercises, or education was prescribed **Duration of therapy period**: 8 sessions, 4 weeks **Duration of follow-up**: 4 weeks	**Pain** (NRS101, 0–100) Baseline mean: Group A 52.5, Group B 48.0 End of study mean: Group A 23.5, Group B 18.7 Absolute benefit: Group A 29.0, Group B 29.3 Reported results: no significant difference SMD: 0.29 (95%CI: −0.43 to 1.01) [power 99%] **Function** (Neck Disability Index, 0–50) Baseline mean: Group A 31.8, Group B26.8 End of study mean: Group A 13.5, Group B 11.0 Absolute benefit: Group A 18.3, Group B 15.8 SMD: 0.23 (95%CI: −0.48 to 0.95) [power 96%] **Patient satisfaction**: NR **Side effects**: NR **Cost of care**: NR
Yurkiw and Mior (1996) Subacute mechanical neck disorder n(A/R) 28/28	**Index treatment** Mechanically assisted device (MAD): frequency: manipulation using Activator Adjusting Instrument, Technique described by Petterson, patient in prone position, instrument in the "2 ring" setting, applied to posterior pillar of the restricted lower cervical vertebrae, one click application was given; frequency: 1 session; route: cervical spine **Comparison**: Spinal Manipulation Therapy (SMT): technique: manipulation, high velocity, low amplitude manual adjustment – Diversified techniques, applied one technique to lower cervical vertebrae (C3–C7); frequency 1 session; route: cervical spine **Co-intervention**: avoided **Duration of therapy period**: 1 session **Duration of follow-up**: none	**Pain** (VAS, 10 cm) Baseline mean: MAD 32.9, SMT 32.9 End of study mean: MAD 20.4, SMT 21.9 Absolute benefit: MAD 12.5, SMT 11.0 Reported results: no significant difference SMD: −0.07 (95%CI: −0.81 to 0.67) [power 61%] **Function**: NR **Patient satisfaction**: NR **GPE**: NR **QoL**: NR **Side effects**: NR **Cost of care**: NR

n (*A*/*R*) – sample number analyzed/randomised; v – versus; NR – not reported; VAS – visual analogue scale; NRS – numeric rating scale; NDI – Neck Disability Index; SMD – standard mean difference, GPE – global perceived effect; QoL – quality of life; RR – relative risk; CI – confidence interval; *p* – probability value; ANOVA – analysis of variance; MONOVA – multiple analysis of variance; s –sessions.

of treatments, and cost of care. All results reported were based on the sample size analysed using 'intention-to-treat' principle, in other words, the sample entering the study.

For continuous data, we calculated standardised mean differences with 95% confidence intervals (SMD; 95%CI). SMD was used because different measures are frequently used to address the same clinical outcome. The Cochrane Back Review Group guidelines (Furlan et al., 2009) were foundational to key estimations of minimum clinically important differences for pain and function/disability. We assumed the minimum clinically important difference to be 10 on a 100-point pain intensity scale (Goldsmith et al., 1993; Felson et al., 1995; Farrar et al., 2001). Similarly, we judged a minimum clinically important difference of 5/50 Neck Disability Index (NDI) units or 10% to be relevant for the NDI (Stratford et al., 1999); a recent systematic review (MacDermid et al., 2009) reports the minimal detectable change to vary from 5/50 for non-complicated neck pain up to 10/50 for cervical radiculopathy. The clinically important difference ranges from 5/50 to 19/50 and was noted to be inconsistent across different studies. For other outcomes (i.e. GPE and QoL scales) where there is an absence of clear guidelines on the size of a clinically important effect sizes, we used a commonly applied system by Cohen (1988): small (0.20), medium (0.50) or large (0.80).

For continuous outcomes reported as medians, we calculated effect sizes (Kendal and Stuart, 1963, p. 237). We calculated relative risks (RR) for dichotomous outcomes. An RR less than one represented a beneficial treatment. We calculated the number needed to treat (NNT: the number of patients a clinician needs to treat in order to achieve a clinically important improvement in one) and treatment advantages (%: the clinically important difference or change in percent) for primary findings, to give the reader a sense of the magnitude of the treatment effect (see Table 2 for operational definition, calculations, results). Power analyses were conducted for each article reporting non-significant findings (Dupont and Plummer, 1990).

Prior to calculation of a pooled effect measure, we assessed the reasonableness of pooling on clinical grounds. We tested the statistical heterogeneity ($P > 0.1$ and $I^2 > 40$) between the studies using a random-effects model.

3.3.4. Qualitative analysis of trial results

We assessed the quality of the body of the evidence using the GRADE approach (Higgins and Green 2008; Furlan et al., 2009;

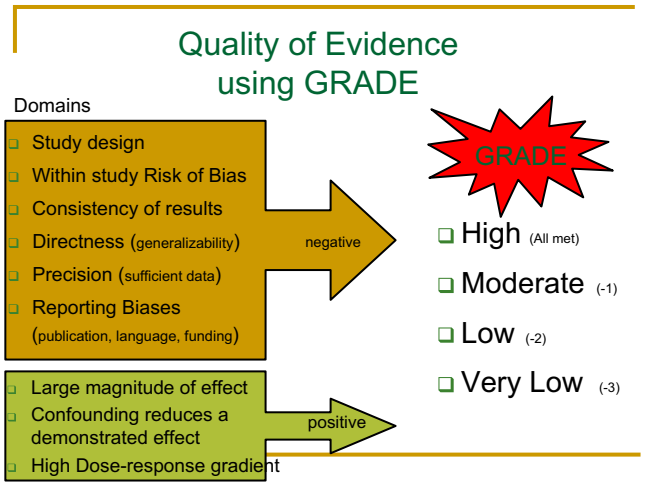


Fig. 1. Quality of evidence using grade.

see Gross et al., 2010 for details; Fig. 1; and Appendix 3). ***High quality evidence***: Further research is very unlikely to change our confidence in the estimate of effect. There are consistent findings among 75% of RCTs with low risk of bias that are generalizable to the population in question. There are sufficient data, with narrow CI. There are no known or suspected reporting biases. ***Moderate quality evidence***: Further research is likely to have an important impact on our confidence in the estimate of effect and may change the estimate. ***Low quality evidence***: Further research is very likely to have an important impact on our confidence in the estimate of effect and is likely to change the estimate. (Two of the domains are not met.) ***Very low quality evidence***: We are very uncertain about the estimate. (Three of the domains are not met.) ***No evidence***: no RCTs were identified that measured the outcome.

4. Results

4.1. Description of studies

Fig. 2 and Gross et al. (2010) describe the flow of the studies from our previous updates (1011 citation postings) and this update (809 citation postings). Of 68 identified RCTs representing 114 publications, we selected 27 RCTs (1522/1805 participants analysed/randomised) representing 32 publications for manipulation or mobilisation performed as a single-modal application. All included trials were small, with fewer than 70 subjects per intervention arm. Agreement between pairs of independent review authors from diverse professional backgrounds for manual therapy was Kw 0.83, SD 0.15.

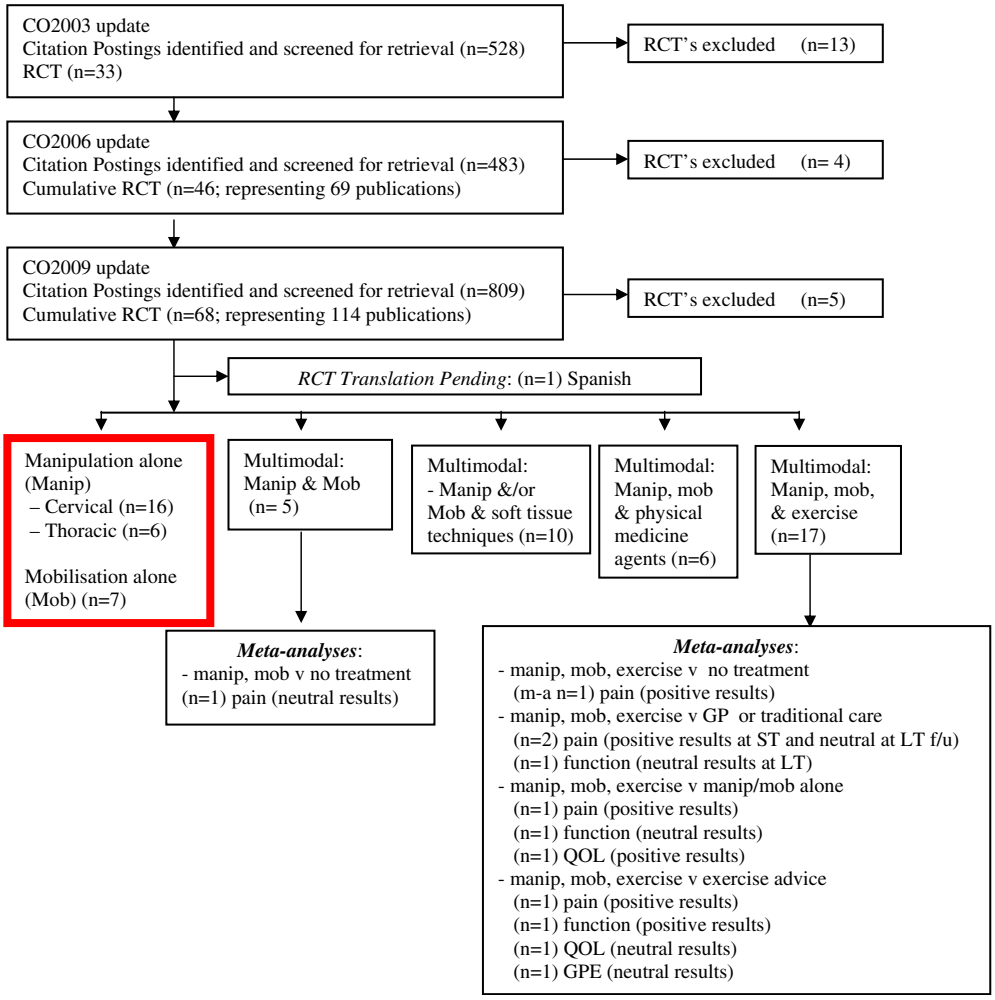


Fig. 2. Flow diagram of study selection.

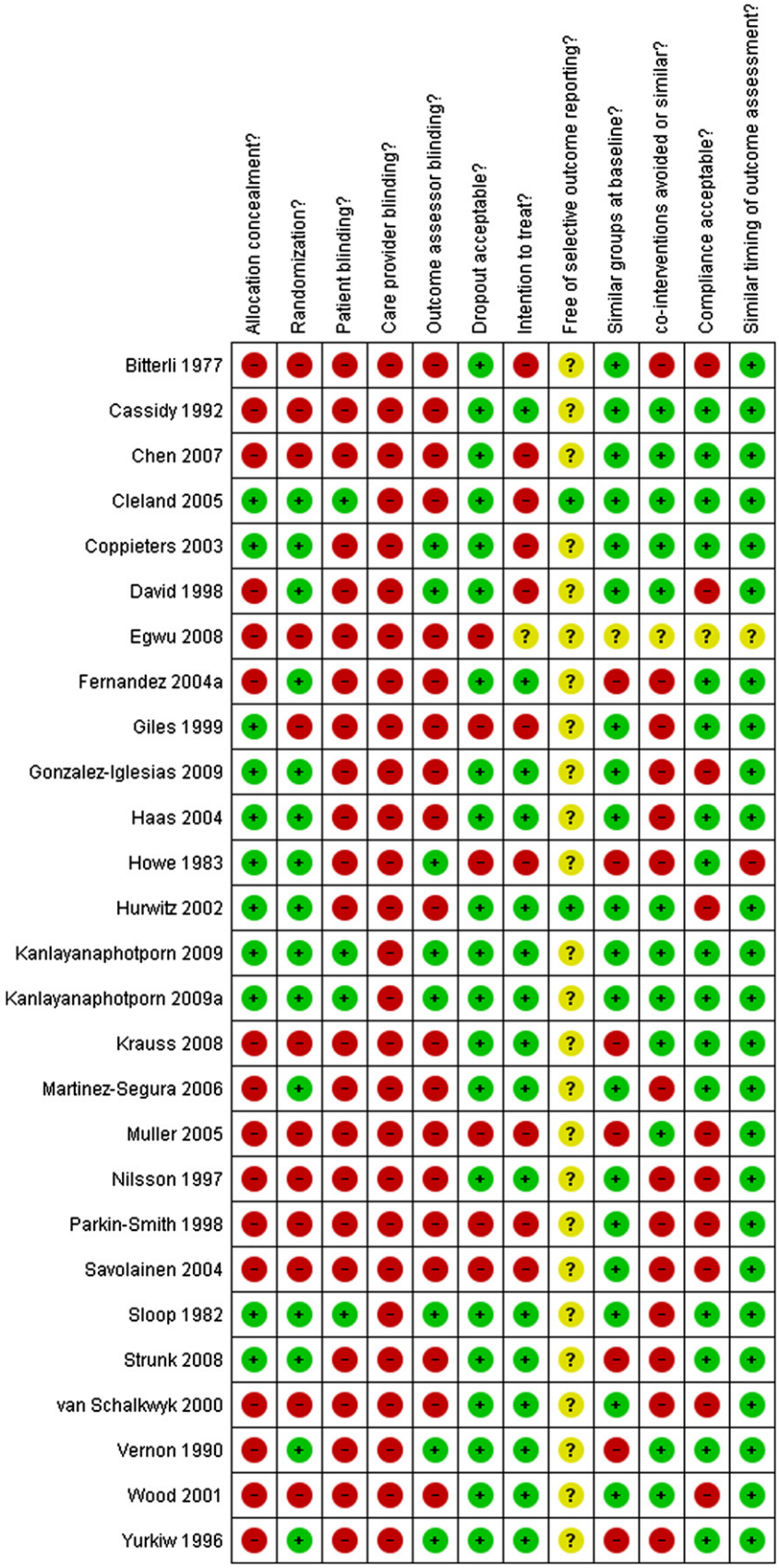


Fig. 3. Methodological quality summary: review of authors' judgments about each methodological quality item for each included study.

4.2. Risk of bias in included studies

See Fig. 3 for summary table of risk of bias findings. Nine of 27 studies had a low risk of bias. We found common risks of bias in the included studies to be: failure to describe or use appropriate concealment of allocation (59%, 16/27) and lack of effective blinding procedures (observer 55% (15/27); patient 81% (22/27); care provider 100% (27/27)). We acknowledge that it is difficult to blind the patient and impossible to blind the care provider in manual treatments. Co-intervention was avoided in only a small number of studies (37%; 10/27) and compliance monitored in 51% (14/27). We do not believe that risk of bias influenced the conclusions, however, we were unable to formally test this notion using meta-regression because we did not have enough data in any one disorder and treatment category.

4.3. Effects of interventions

4.3.1. Manipulation alone of cervical region

Sixteen trials met the inclusion criteria for this section. Four of these studies had a low risk of bias (Sloop et al., 1982; Hurwitz et al., 2002; Haas et al., 2004; Strunk and Hondras, 2008) and 12 had a high risk of bias (Bitterli et al., 1977; Howe et al., 1983; Vernon et al., 1990; Cassidy et al., 1992; Nilsson, 1995*; Nilsson et al., 1996*, 1997*; Yurkiw and Mior, 1996; Giles and Muller, 1999; van Schalkwyk and Parkin-Smith, 2000; Wood et al., 2001; Muller and Giles, 2005; Martinez-Segura et al., 2006; Chen et al., 2007). Asterisk denotes multiple publications from one trial.

There is moderate quality evidence (two trials, 369 participants; Cassidy et al., 1992; Hurwitz et al., 2002; see Fig. 4) that manipulation produces similar changes in pain, function and patient satisfaction when compared to mobilisation for subacute or chronic neck pain at short- and intermediate-term follow-up.

There is low quality evidence (three trials, 130 participants; Bitterli et al., 1977; Sloop et al., 1982; Martinez-Segura et al., 2006; see Fig. 5) that manipulation alone versus a control may provide immediate- and short-term pain relief following one to four treatment sessions in subjects with acute or chronic neck pain. Low quality evidence from one small dosage trial (25 participants; Haas et al., 2004) suggests that nine or 12 sessions of manipulation are superior to three for pain relief at immediate post-treatment follow-up and neck-related disability for chronic cervicogenic headache. Larger dose-finding trials are needed to establish the optimal dose.

There is very low quality evidence at short-term follow-up that:

- one manipulation technique is not superior to another for pain reduction for subacute neck pain (three trials, 88 participants; Yurkiw and Mior, 1996; van Schalkwyk and Parkin-Smith, 2000; Wood et al., 2001);
- manipulation is equivalent to certain medication (2 trials, 69 participants; Sloop et al., 1982; Howe et al., 1983), acupuncture (2 trials, 81 participants; Giles and Muller, 1999; Muller and Giles, 2005), certain soft-tissue treatments (1 trial, 53 participants; Nilsson et al., 1997*) or certain combined treatments for subacute and chronic neck pain and to some extent improved function (Strunk and Hondras, 2008); and
- manipulation may be superior to TENS (1 trial, 64 participants; Chen et al., 2007) for individuals with chronic cervicogenic headache.

4.3.2. Manipulation alone of thoracic region

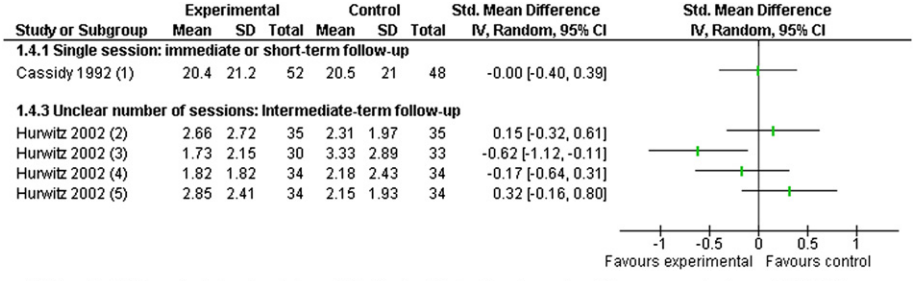
Six trials, one with a low risk of bias (Cleland et al., 2005) and five with a high risk of bias (Parkin-Smith and Penter, 1998; Fernandez-de-las-Penas et al., 2004; Savolainen et al., 2004; Krauss et al., 2008; Gonzalez-Iglesias et al., 2009) met the inclusion criteria for this section (see Fig. 6 for continuous pain data).

Low quality of evidence (one trial, 36 participants, Cleland et al., 2005) favoured a single session of thoracic manipulation for immediate pain reduction when compared to a placebo for chronic neck pain (NNT 5, 29% treatment advantage). There is low quality evidence (one trial, 45 participants; Gonzalez-Iglesias et al., 2009) in support of thoracic manipulation as an adjunct to electrothermal therapy for both pain (NNT 7; 46.6% treatment advantage) and function (NNT 5; 40.6% treatment advantage) for acute neck pain. Very low quality evidence (one trial, 88 participants; Fernandez-de-las-Penas et al., 2004) supports thoracic manipulation as an adjunct to individualised physiotherapy care for pain reduction for acute/subacute WAD. Alternatively, very low quality evidence found no difference in pain during neck rotation or function when a thoracic manipulation was compared to no treatment (one trial, 30 participants; Krauss et al., 2008) or exercise (one trial, 41 participants; Savolainen et al., 2004) for neck pain of undefined duration. When thoracic manipulation was added to cervical manipulation alone, there was very low quality evidence (Parkin-Smith and Penter, 1998) suggesting no added benefit for participants with neck pain of undefined duration.

4.3.3. Mobilisation alone of cervical region

Eight trials, five with a low risk of bias (David et al., 1998; Coppieters et al., 2002*,2003*; Hurwitz et al., 2002; Kanlayanaphotporn et al., 2009, 2010) and three with a high risk of bias (Bitterli et al., 1977; Cassidy et al., 1992; Egwu, 2008) met the inclusion criteria (see Fig. 7 for continuous pain data).

One factorial design trial (1 RCT with 4 independent comparisons, 268 participants; Hurwitz et al., 2002) and one comparison trial (1 trial, 100 participants; Cassidy et al., 1992) provided



(1) Cassidy 1992: manipulation (manip) v mobilisation (mob); duration: 1 session; follow-up: none; instrument: NRS 101
(2) Hurwitz 2002: manipulation v mobilisation; duration: NR; follow up: 6 month; instrument: NRS (0 to 10)
(3) Hurwitz 2002: manipulation and heat and EMS v mobilisation and heat and EMS; duration: NR; follow up: 6 month; instrument:
(4) Hurwitz 2002: manipulation and heat v mobilisation and heat; duration: NR; follow up: 6 month; instrument: NRS (0 to 10)
(5) Hurwitz 2002: manipulation and EMS v mobilisation and EMS; duration: NR; follow up: 6 month; instrument: NRS (0 to 10)

Fig. 4. Manipulation versus mobilisation for pain at short- and intermediate-term follow-up demonstrates no difference.

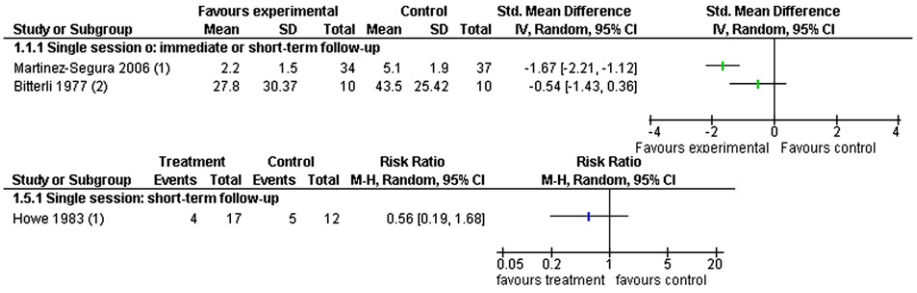


Fig. 5. Manipulation versus a control for pain at immediate- and short-term follow-up.

moderate to low quality evidence showing no difference between mobilisation compared to manipulation and other treatments for pain, function and patient satisfaction for subacute/chronic neck pain. There was low quality evidence (1 trial, 51 participants; David et al., 1998) that a combination of Maitland mobilisation techniques was as effective as acupuncture for pain relief and improved function immediately post-treatment for subacute/chronic neck pain. Low quality evidence exists from one small trial (20 participants; Coppieters et al., 2003*) suggesting neural dynamic techniques led to statistically insignificant, but clinically important pain reduction immediately post-treatment in participants with neck pain of mixed duration.

Certain mobilisation techniques may be superior to others. Very low quality evidence (one trial, 95 participants; Egwu, 2008) suggests that an anterior–posterior mobilisation was more effective than a transverse oscillatory and rotational mobilisation immediately post-treatment for acute/subacute neck pain. Low quality evidence (two trials, 120 participants; Egwu, 2008) suggests no difference in pain reduction or GPE for chronic neck pain when comparing an ipsilateral posterior–anterior or central posterior–anterior to one of three random posterior–anterior mobilisations.

4.4. Other considerations

4.4.1. Adverse events

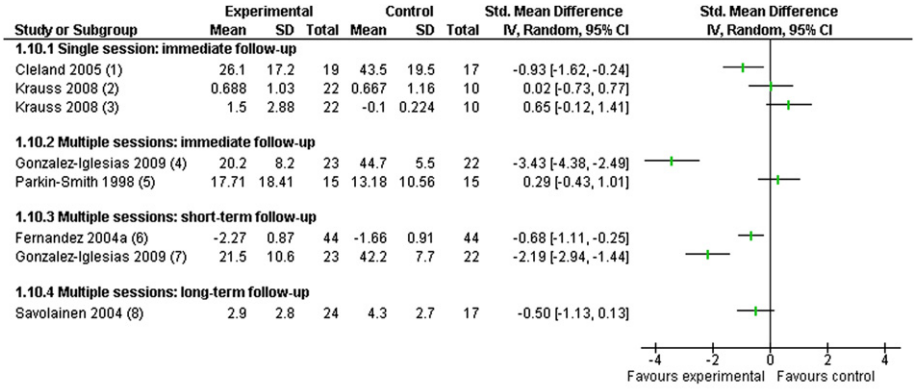
The number of subjects experiencing side effects was reported in 29% (8/27) of trials. Three out of those eight trials reported no side effects and five trials reported benign and transient side effects including radicular symptoms (58/1000 in the treatment group versus 20/1000 in the control group) headache or neck pain (105/1000 in the treatment group versus 52/1000 in the control group) (Carlesso et al., in press). The rate of rare but serious adverse events such as strokes or serious neurological deficits could not be established from our review.

4.4.2. Cost of care

No direct measures of the cost of care were reported in the 27 trials included in this review.

5. Discussion

This update investigates the effect of mobilisation or manipulation alone in the treatment of neck pain and is a co-publication of a Cochrane Review. The overall methodological quality of trials has not improved in spite of CONSORT guidelines (Schultz et al., 2010). The evidence suggests some immediate- or short-term pain relief with a course of cervical manipulation or mobilisation alone, but these benefits are not maintained over the long-term. Evidence seems to support the use of thoracic manipulation alone for immediate pain relief or as an adjunct to electrothermal or individualised physiotherapy treatment strategies for pain relief and improvements in function. One dose–response trial and one trial with a low risk of bias that used a factorial design emerged. Certain mobilisation techniques may be superior.



(1) Cleland 2005: manipulation v placebo; duration: 1 session; follow up: none; instrument: VAS (0 to100)
(2) Krauss 2008: manipulation (TSM) v no treatment; duration:1 session; follow up: none; instrument: 9 point faces scale *
(3) Krauss 2008: manipulation (TSM) v no treatment: duration:1 session; follow up: none; instrument: 9 point faces scale**
(4) Gonzalez-Isglesias 2009: manipulation v ETT both arms; duration: 3 weeks, 5 sessions; follow up: none; instrument: VAS (0 to100)
(5) Parkin-Smith 1998: T/S manipulation v C/S manipulation both arms; duration: 3 weeks, 6 sessions; follow up: none; instrument:NRS101
(6) Fernandez 2004a: manipulation v PT both arms; duration: 2 weeks, 15 sessions; follow up: none; instrument: VAS (0-10)
(7) Gonzalez-Iglesias 2009: manipulation v ETT both arms; duration: 3 weeks, 5 sessions; follow up: 4 weeks; instrument: VAS (0 to 100)
(8) Savolainen 2004: manipulation v instructed exercise; duration: NR; follow up 52 weeks; instrument: VAS - current pain (0 to 10)

Fig. 6. Manipulation of the thoracic region for pain outcomes.

Table 2
NNT & treatment advantage.

Author/comparison	NNT	Treatment advantage (%)
Martinez-Segura et al. (2006), outcome: pain	2 [Clinically important pain reduction]	54
Cleland et al. (2005), outcome: pain	5 [Clinically important pain reduction]	29
Fernandez-de-la-Penas et al. (2004), outcome: pain	Unable to calculate due to baseline data not reported (author was unable to provide these data)	
Gonzalez-Iglesias et al. (2009), outcome: pain	7 [Clinically important pain reduction]	46.6
Gonzalez-Iglesias et al. (2009), outcome: function	5 [Clinically important functional improvements]	40.6
Martinez-Segura et al. (2006), outcome: pain	2 [Clinically important pain reduction]	54.0

% Treatment advantage: Calculation of the clinically important difference or change on a percent scale was estimated as follows. Karlberg et al., (1996) data are used in this example: The assumption made was that a positive mean/median value is improvement and a negative is deterioration. Treatment mean/median (SD) at baseline 54(23) and at final (SD) 31(10); Control mean/median (SD) at baseline 56(1) and at final 55(20). The % improvement [treatment] equals the difference between the change in the treatment group [23] divided by the treatment baseline [54] which equals 42.6%. The % improvement [control] equals the difference between the change in the control group [1] divided by the control baseline [56] which equals 1.8%. The treatment advantage equals 42.6% minus 1.8% which equals 40.8%.
Number needed to treat (NNT): The number of patients a clinician needs to treat in order to achieve a clinically important improvement in one. Assuming the minimal clinically important difference to be 10%, the baseline of the experimental mean of 54 minus 10 units change on a 100-point scale equals 44 and for the control 56 minus 10 units change equals 46. For experimental group: 44 minus 31 divided by 10 equals $z = 1.3$ which gives an area under the normal curve of 0.9032. For control group: 46 minus 55 divided by 20 equals $z = -0.5$ which gives an area under the normal curve of 0.3085. Then 0.9032 minus 0.3085 equals 0.5947. NNT equals 1 divided by 0.5947 which equals 1.66 or 2 when rounded.

For WAD, our findings differed slightly from Verhagen et al. (2007) because they did not include manipulation of the thoracic spine as a passive treatment of neck pain. We are in concordance with their other findings showing a dearth of evidence for mobilisation or manipulation for WAD. For acute neck disorder that is not whiplash-associated, we also agreed with Vernon et al. (2005), who noted a striking lack of high quality evidence. For chronic non-specific neck pain, we agreed with Brønfort et al. (2004) that trials were sparse and inconclusive for the application of manipulation or mobilisation as a single approach. We were also in agreement with Hurwitz et al. (2008), who found evidence of short-term benefit for mobilisation in comparison to usual GP care, pain medications, or advice to stay active. Our findings were in disagreement with Gemmel and Miller (2006), who described five trials with a high risk of bias that showed mixed findings and concluded that no one therapy was more effective than the others. We disagreed with Vernon et al. (2007), who indicated moderate to high quality evidence in support of spinal manipulation or mobilisation for chronic non-specific neck pain. This discordance was not based on new literature but was related to the framing of the systematic review's question and methodology used. Vernon et al. (2007) included cohort studies while our review solely utilized RCT data; they did not calculate SMD from the RCT published data but rather noted the reported direction of effect; the data were not always consistent with the reported direction of effect. There were also differences in interpretation; we suggested that one treatment with only immediate post-treatment outcomes did not denote effectiveness, did not reflect clinical practice and certainly did not give us any indication of either short- or long-term effects. Thus, we suggested the evidence remained unclear. Overall, differences in findings stemmed from differing definitions or clustering of treatments; from differing disorder subgroup classifications; and from different reported time frames for outcome measures.

How do we know when mobilisations will be effective? It's still not clear. Meta-analyses, subgroup analyses, and sensitivity analyses were hampered by the wide spectrum of comparisons, treatment characteristics and dosages. Until the quantity of quality studies increases within individual subgroups of neck pain, we will not be able to make any strong conclusions about which groups benefit most from manipulation or mobilisation. Cleland et al. (2007) developed clinical prediction rules for the

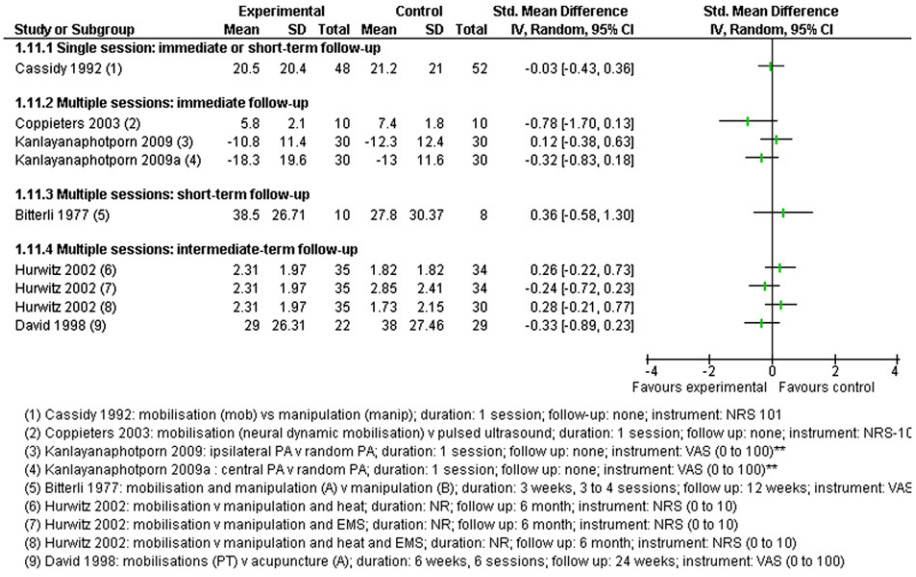


Fig. 7. Manipulation of the cervical region for pain at immediate-, short-term and intermediate-term follow-up.

use of thoracic manipulation in the treatment of neck pain. Similar studies investigating clinical prediction rules for other mobilisations or manipulations in varied neck pain populations would help clinicians to determine when to utilize manual therapy.

What is the optimal "dosage" or are the optimal "clinical parameters" for a given technique category? We don't know. From the data on pain relief we have to date, the most commonly reported factors were frequency (total number of sessions) and duration (total number of weeks). The ideal dosage for cervical manipulation, thoracic manipulation, or cervical mobilisation for the treatment of cervical pain could not be determined when evaluating existing controlled trials. Pilot studies of mobilisation and manipulation exploring the minimally effective dose as well as the optimal dose should be conducted prior to doing a larger trial. These pilot studies would serve a purpose similar to the small dose-finding studies conducted in pharmaceutical trials that are used to establish a minimally effective dose.

Adverse events reported from RCTs in this review were benign, transient side effects. Clearly, smaller randomised trials are unlikely to detect rare adverse events. From surveys and review articles, the risk of a serious irreversible complication (e.g., stroke) for cervical manipulations has been reported to vary from one adverse event in 3020 to one in 1,000,000 manipulations (Assendelft et al., 1996; Gross et al., 2002b). Better reporting of adverse events is required. Additionally, new trials are necessary to determine whether there is an economical advantage in using manipulation or mobilisation techniques to treat neck pain.

We have observed four positive advances in recent years. Some trials were larger, had a lower risk of bias, had longer-term follow-up, and used self-reported ratings (e.g., pain, disability self-report questionnaires, and GPE) as primary outcomes on a more consistent basis. Access Gross et al. (2010) for a detailed discussion on key methodological issues. One overarching need remains, that is, adherence to and consistent reporting using CONSORT guidelines (Schultz et al., 2010).

A total of 14 ongoing studies have been identified in this review from 1995 through 2006. We are tracking three of these studies, but the remaining 11 studies were not located through our literature search. With such a large number of study protocols and pilot studies not reaching publication, we question if a publication bias is leading towards publication of primarily positive results.

6. Conclusion

6.1. Implications for practice

Manipulation for acute to chronic neck pain:

- cervical manipulation produces similar pain relief, functional improvements, and patient satisfaction to mobilisation
- cervical manipulation may provide short-term, but not long-term pain relief
- thoracic manipulation used alone or in combination with electrothermal or individualised physiotherapy treatments may improve pain and function

Mobilisation for neck pain of mixed duration:

- cervical mobilisation is similar to manipulation or acupuncture for pain and function
- one mobilisation technique may be superior to another (that is, anterior–posterior mobilisation may be superior to transverse oscillatory and rotational mobilisation; ipsilateral posterior–anterior or central posterior–anterior may be better then one of three random posterior–anterior mobilisations).

6.2. Implications for research

Meta-analysis of data across trials and sensitivity analysis were hampered by the wide spectrum of comparisons, treatment characteristics and dosages. Factorial design would help to determine the active treatment agent(s) within a treatment mix. Phase II trials would help to identify the most effective treatment characteristics and dosages. Greater attention to methodological quality continues to be needed.

Acknowledgements

We are indebted to the many authors of primary studies for their support in retrieving original research. We thank our volunteers, translators, students, the Cochrane Back Review Group editors, and Lisa Carlesso and Colleen McPhee for early access to their research.

Declarations of interest: Two of our review authors are authors of included studies. Although Gert Brønfort and Jan Hoving were review authors, they were not involved in decisions about the inclusion, risk of bias assessment or data extraction of their studies.

Appendix 1. MEDLINE search strategy

1. neck/ or neck muscles/ or exp cervical plexus/ or exp cervical vertebrae/ or atlanto-Axial Joint/ or atlanto-occipital joint/ or axis/ or atlas/ or spinal nerve roots/ or exp brachial plexus/
2. (odontoid or cervical or occip: or atlant:).tw.
3. 1 or 2
4. exp arthritis/ or exp myofascial pain syndromes/ or fibromyalgia/ or spondylitis/ or exp spinal osteophytosis/ or spondylolisthesis/
5. exp headache/ and cervic:.tw.
6. whiplash injuries/ or cervical rib syndrome/ or torticollis/ or cervico-brachial neuralgia.ti,ab,sh. or exp radiculitis/ or polyradiculitis/ or polyradiculoneuritis/ or thoracic outlet syndrome/
7. (monoradicul: or monoradicl:).tw.
8. 4 or 5 or 6 or 7
9. random:.ti,ab,sh.
10. randomised controlled trial.pt.
11. double-blind method/
12. single blind method/
13. placebos/
14. clinical trial.pt.
15. exp clinical trials/
16. controlled clinical trial.pt.
17. (clin$ adj25 trial$).ti,ab.
18. ((singl$ or doubl$ or trebl$) adj25 (blind$ or mask$)).ti,ab.
19. placebo$.ti,ab.
20. or/9–19
21. exp arthritis/rh,th or exp myofascial pain syndromes/rh,th or fibromyalgia/rh,th or spondylitis/rh,th or exp spinal osteophytosis/rh,th or spondylosis/rh,th or spondylolisthesis/rh,th
22. exp headache/rh,th and cervic:.tw.
23. whiplash injuries/rh,th or cervical rib syndrome/rh,th or thoracic outlet syndrome/rh,th or torticollis/rh,th or cervico-brachial neuralgia/rh,th or exp radiculitis/rh,th or polyradiculitis/rh,th or polyradiculoneuritis/rh,th

24. or/21–23
25. exp alternative medicine/ or chiropractic/
26. (acupuncture or biofeedback or chiropract: or electric stimulation therapy or kinesiology or massage or traditional medicine or relaxation or therapeutic touch).tw.
27. or/25–26
28. 3 and 24
29. 3 and 8 and 27
30. 28 or 29
31. 20 and 30

Appendix 2. Criteria for a judgment of yes for the sources of risk of bias

Assessing risk of bias in relation to adequate or inadequate sequence generation.

1. Was the method of randomisation adequate?

A random (unpredictable) assignment sequence. Examples of adequate methods are coin toss (for studies with two groups), rolling a dice (for studies with two or more groups), drawing of balls of different colours, drawing of ballots with the study group labels from a dark bag, computer-generated random sequence, pre-ordered sealed envelops, sequentially-ordered vials, telephone call to a central office, and pre-ordered list of treatment assignments

Examples of inadequate methods are: alternation, birth date, social insurance/security number, date in which they are invited to participate in the study, and hospital registration number

2. Was the treatment allocation concealed?

Assignment generated by an independent person not responsible for determining the eligibility of the patients. This person has no information about the persons included in the trial and has no influence on the assignment sequence or on the decision about eligibility of the patient.

Was knowledge of the allocated interventions adequately prevented during the study?

3. Was the patient blinded to the intervention?

This item should be scored "yes" if the index and control groups are indistinguishable for the patients or if the success of blinding was tested among the patients and it was successful.

4. Was the care provider blinded to the intervention?

This item should be scored "yes" if the index and control groups are indistinguishable for the care providers or if the success of blinding was tested among the care providers and it was successful

5. Was the outcome assessor blinded to the intervention?

Adequacy of blinding should be assessed for the primary outcomes. This item should be scored "yes" if the success of blinding was tested among the outcome assessors and it was successful or:

- for patient-reported outcomes in which the patient is the outcome assessor (e.g., pain, disability): the blinding procedure is adequate for outcome assessors if participant blinding is scored "yes"
- for outcome criteria assessed during scheduled visit and that supposes a contact between participants and outcome assessors (e.g., clinical examination): the blinding procedure is adequate if patients are blinded, and the treatment or adverse effects of the treatment cannot be noticed during clinical examination
- for outcome criteria that do not suppose a contact with participants (e.g., radiography, magnetic resonance imaging): the blinding procedure is adequate if the treatment or adverse effects of the treatment cannot be noticed when assessing the main outcome
- for outcome criteria that are clinical or therapeutic events that will be determined by the interaction between patients and care providers (e.g., co-interventions, hospitalization length, treatment failure), in which the care provider is the outcome assessor: the blinding procedure is adequate for outcome assessors if the item for care providers is scored "yes"
- for outcome criteria that are assessed from data of the medical forms: the blinding procedure is adequate if the treatment or adverse effects of the treatment cannot be noticed on the extracted data

Were incomplete outcome data adequately addressed?

6. Was the drop-out rate described and acceptable?

The number of participants who were included in the study but did not complete the observation period or were not included in the analysis must be described and reasons given. If the percentage of withdrawals and drop-outs does not exceed 20% for short-term follow-up and 30% for long-term follow-up and does not lead to substantial bias a 'yes' is scored. (N.B. these percentages are arbitrary, not supported by literature).

7. Were all randomised participants analysed in the group to which they were allocated?

All randomised patients are reported/analysed in the group they were allocated to by randomisation for the most important moments of effect measurement (minus missing values) irrespective of non-compliance and co-interventions.

8. Are reports of the study free of suggestion of selective outcome reporting?

In order to receive a 'yes', the review author determines if all the results from all pre-specified outcomes have been adequately reported in the published report of the trial. This information is either obtained by comparing the protocol and the report, or in the absence of the protocol, assessing that the published report includes enough information to make this judgment.

Other sources of potential bias:

9. Were the groups similar at baseline regarding the most important prognostic indicators?

In order to receive a "yes", groups have to be similar at baseline regarding demographic factors, duration and severity of complaints, percentage of patients with neurological symptoms, and value of main outcome measure(s).

10. Were co-interventions avoided or similar?

This item should be scored "yes" if there were no co-interventions or they were similar between the index and control groups.

11. Was the compliance acceptable in all groups?

The reviewer determines if the compliance with the interventions is acceptable, based on the reported intensity, duration,

number and frequency of sessions for both the index intervention and control intervention(s). For example, physiotherapy treatment is usually administered over several sessions; therefore it is necessary to assess how many sessions each patient attended. For single-session interventions (for example: surgery), this item is irrelevant.

12. Was the timing of the outcome assessment similar in all groups?

Timing of outcome assessment should be identical for all intervention groups and for all important outcome assessments. Note: These instructions are adapted from van Tulder (2003) and the Cochrane Handbook 2008 of Systematic Reviews of Interventions.

Appendix 3. Grading the quality of evidence – definition of domains

Study Design refers to type of study (i.e. randomised, observational study)

Limitations in Design (Quality) refers to the 12 risk of bias criteria noted in Appendix 2.

Consistency refers to the similarity of results across studies. When all studies are included in the meta-analysis, 'consistency' is defined as absence of statistical heterogeneity. In the case that not all studies are combined in a meta-analysis, 'consistency' is defined when all studies for the specific outcome lead to the same decision or recommendation, and 'inconsistency' is present if the results of two or more studies lead to clinically different decisions or recommendations. Review authors use their judgment to decide if there is inconsistency when only one study leads to clinically different decision or recommendation.

Directness (generalizability) refers to the extent to which the people, interventions and outcome measures are similar to those of interest.

Precision of the evidence relates to the number of studies, patients and events for each outcome. Imprecise data are defined as:

- Only one study for an outcome, regardless of the sample size or the CI
- Multiple studies combined in a meta-analysis: the CI is sufficiently wide that the estimate is consistent with conflicting recommendations. For rare events one should consider the CI around the risk difference rather than the CI around the RR
- Multiple studies not combined in a meta-analysis: the total sample size is underpowered to detect a clinically significant difference between those who received the index intervention compared to those who received the control intervention. In this case, a post-hoc sample size calculation should be performed to determine the adequate sample size for each outcome

Reporting (Publication) bias should only be considered present if there is actual evidence of reporting bias rather than only speculation about reporting bias. The Cochrane Reporting Bias Methods Group describes the following types of reporting bias and definitions:

- Publication Bias: the publication or non publication of research findings, depending on the nature and direction of the results.
- Time Lag Bias: the rapid or delayed publication of research findings, depending on the nature and direction of the results.
- Language Bias: the publication of research findings in a particular language, depending on the nature and direction of the results.
- Funding Bias: the reporting of research findings, depending on how the results accord with the aspirations of the funding body.
- Outcome Variable Selection Bias: the selective reporting of some outcomes but not others, depending on the nature and direction of the research findings.
- Developed Country Biases: the non publication or non indication of findings, depending on whether the authors were based in developed or in developing countries.

References

Assendelft WJ, Bouter LM, Knipschild PG. Complications of spinal manipulation: a comprehensive review of the literature. Journal of Family Practice 1996;42(5):475–80.

Beattie P. Measurement of health outcomes in the clinical setting: applications to physiotherapy. Physiotherapy Theory and Practice 2001;17:173–85.

Bitterli J, Graf R, Robert F, Adler R, Mumenthaler M. Zur objectivierung der manualtherapeutischen beeinflussbarkeit des spondylogenen kopfschmerzes. Nervenarzt 1977;48:259–62.

Bodgduk N. Whiplash: why pay for what does not work? Journal of Musculoskeletal Pain 2001;8(1/2):29–53.

Borghouts JAJ, Koes BW, Bouter LM. The clinical course and prognostic factors of non-specific neck pain: a systematic review. Pain 1998;77:1–13.

Borghouts JAJ, Koes BW, Bouter LM. Cost-of-illness in neck pain in the Netherlands in 1996. Pain 1999;80:629–36.

Brønfort G, Haas M, Evans RL, Boutar LM. Efficacy of manipulation and mobilization for low back pain and neck pain: a systematic review and best evidence synthesis. The Spine Journal 2004;4(3):335–56.

Cassidy JD, Lopes AA, Yong-Hing K. The immediate effect of manipulation versus mobilization on pain and range of motion in the cervical spine: a randomized controlled trial. Journal of Manipulative and Physiological Therapeutics 1992;15(9):570–5.

Carlesso L, Gross A, Santaguida PL, Burnie S, Voth S, Sadi J. Adverse events associated with the use of cervical manipulation and for the treatment of neck pain in adults: a systematic review. Manual Therapy, in press, doi:10.1016/j.math.2010.02.006.

Cicchetti DV. Assessing inter-rater reliability for rating scales: resolving some basic issues. British Journal of Psychiatry 1976;129:452–6.

Chen L, Zhang XL, Ding H, Tao YQ, Zhan HS. Comparative study on effects of manipulation treatment and transcutaneous electrical nerve stimulation on patients with cervicogenic headache. Journal of Chinese Integrative Medicine 2007;5(4):403–6.

Cleland JA, Childs JD, McRae M, Palmer JA, Stowell T. Immediate effects of a thoracic manipulation in patients with neck pain: a randomized clinical trial. Manual Therapy 2005;10:127–35.

Cleland JA, Childs JD, Fritz JM, Whitman JM, Eberhart SL. Development of a clinical prediction rule for guiding treatment of a subgroup of patients with neck pain: use of thoracic spine manipulation, exercise, and patient education. Physical Therapy 2007;87(1):9–23.

Cohen J. Statistical power analysis for the behavioural sciences. 2nd ed. Hillsdale, New Jersey: Lawrence Erlbaum Associates; 1988.

Coppieters MW, Stappaerts KH, Wouters LL, Janssens K. Aberrant protective force generation during neural provocation testing and the effect of treatment in patients with neurogenic cervicogenic pain. Journal of Manipulative and Physiological Therapeutics 2002;26(2):99–106.

Coppieters MW, Stappaerts KH, Wouters LL, Janssens K. The immediate effects of a cervical lateral glide treatment technique in patients with neurogenic cervicobrachial pain. Journal of Orthopedic and Sports Physical Therapy 2003;33:369–78.

Côté P, Cassidy D, Carroll L. The Saskatchewan health and back pain survey. The prevalence of neck pain and related disability in Saskatchewan adults. Spine 1998;23(15):1689–98.

Côté P, Kristman V, Vidmar M, Van Eerd D, Hogg-Johnson S, Beaton D, et al. The prevalence and incidence of work absenteeism involving neck pain: a cohort of Ontario lost-time claimants. Spine 2008;33(4 Suppl.):S192–8.

David J, Modi S, Aluko AA, Robertshaw C, Farebrother J. Chronic neck pain: a comparison of acupuncture treatment and physiotherapy. British Journal of Rheumatology 1998;37:118–22.

D'Sylva J, Gross A, Miller J, Burnie S, Goldsmith C, Graham N, et al. Manual therapy with or without physical medicine modalities: a systematic review. Manual Therapy, in press.

Dupont WD, Plummer WD. Power and sample size calculations: a review and computer program. Controlled Clinical Trials 1990;11:116–8.

Egwu MO. Relative therapeutic efficacy of some vertebral mobilization techniques in the management of unilateral cervical spondylosis: a comparative study. Journal of Physical Therapy and Science 2008;20:103–8.

Farrar JT, Young Jr JP, LaMoureaux L, Worth JL, Poole RM. Clinical importance of changes in chronic pain intensity measured on an 11-point numerical rating scale. Pain 2001;94:149–58.
Felson DT, Anderson JJ, Boers M, et al. American college of rheumatology: preliminary definition of improvement in rheumatoid arthritis. Arthritis and Rheumatism 1995;38(6):727–35.
Fernandez-de-las-Penas C, Fernandez-Carnero J, Plaza Fernandez A, Lomas-Vega R, Miangolarra-Page JC. Dorsal manipulation in whiplash injury treatment: a randomized controlled trial. Journal of Whiplash & Related Disorders 2004;3:55–71.
Finch E, Brooks D, Stratford P, Mayo N. Physical rehabilitation outcome measures: a guide to enhanced clinical decision making. 2nd ed. Philadelphia: Lippincott Williams and Wilkins; 2002.
Furlan AD, Pennick V, Bombardier C, van Tulder M, Editorial Board Cochrane Back Review Group. 2009 Updated method guidelines for systematic reviews in the Cochrane Back Review Group. Spine 2009;34(18):1929–41.
Gemmel H, Miller B. Comparative effectiveness of manipulation, mobilisation and the activator instrument in the treatment of non-specific neck pain: a systematic review. Chiropractic and Osteopathy 2006;14(7):15–21.
Giles LGF, Muller R. Chronic spinal pain syndromes: a clinical pilot trial comparing acupuncture, a nonsteroidal anti-inflammatory drug, and spinal manipulation. Journal of Manipulative and Physiological Therapeutics 1999;22(6):376–81.
Goldsmith CH, Boers M, Bombardier C, Tugwell P. Criteria for clinically important changes in outcomes. Development, scoring and evaluation of rheumatoid arthritis patients and trial profiles. Journal of Rheumatology 1993;20(3):561–5.
Gonzalez-Iglesias J, Fernandez-de-las-Penas C, Cleland JA, Del Rosario Gutierrez-Vega M. Thoracic spine manipulation for the management of patients with neck pain: a randomized clinical trial. Journal of Orthopaedic and Sports Physical Therapy 2009;39(1):20–7.
Gross AR, Aker PD, Goldsmith CH, Peloso P. Conservative management of neck disorders. A systematic overview and meta-analysis. Doc No. 200-201. Online Journal of Current Clinical Trials 1996 [accessed 20.01.02].
Gross AR, Kay T, Hondras M, Goldsmith C, Haines T, Peloso P, et al. Manual therapy for mechanical neck disorders: a systematic review. Manual Therapy 2002a;7(3):131–49.
Gross AR, Kay T, Kennedy C, Gasner D, Hurley L, Yardley K, et al. Clinical practice guideline on the use of manipulation or mobilization in the treatment of adults with mechanical neck disorder. Manual Therapy 2002b;7(4):193–205.
Gross AR, Bronfort G, Hoving J, Goldsmith CH, Kay T, Aker P, et al. Manipulation and mobilisation for mechanical neck disorders. Cochrane Database of Systematic Reviews 2004a;(1):CD004249. doi:10.1002/14651858.CD004249.pub2 [accessed 20.03.10].
Gross AR, Hoving JL, Haines TA, Goldsmith CH, Kay T, Aker P, et al. A Cochrane review of manipulation and mobilization for mechanical neck disorders. Spine 2004b;29(14):1541–8.
Gross AR, Goldsmith C, Hoving JL, Haines T, Peloso P, Aker P, et al. Cervical Overview Group. Conservative management of mechanical neck disorders: a systematic review. Journal of Rheumatology 2007;34(5):1083–1102.
Gross AR, Miller J, D'Sylva J, Burnie SJ, Goldsmith CH, Graham N, et al. Manipulation or mobilisation for neck pain. Cochrane Database of Systematic Reviews 2010;(1):CD004249. doi:10.1002/14651858.CD004249.pub3 [accessed 6.04.10].
Guzman J, Hurwitz EL, Carroll LJ, Haldeman S, Côté P, Carragee EJ, et al. A new conceptual model of neck pain: linking onset, course, and care: the bone and joint decade 2000–2010 task force on neck pain and its associated disorders. Spine 2008;33(4 Suppl.):S14–23.
Haas M, Groupp E, Aickin M, Fairweather A, Ganger B, Attwood M, et al. Dose response for chiropractic care of chronic cervicogenic headache and associated neck pain: a randomized pilot study. Journal Manipulative and Physiological Therapeutics 2004;27:547–53.
Higgins JPT, Green S, editors. Cochrane handbook for systematic reviews of interventions version 5.0.1, www.cochrane-handbook.org; 2008 [accessed 14.11.09].
Headache Classification Committee. The international Calssification of headache disorders. Cephalalgia 2004;24(Suppl. 1):9–160.
Hogg-Johnson S, van der Velde G, Carroll LJ, Holm LW, Cassidy JD, Guzman J, et al. The burden and determinants of neck pain in the general population: results of the bone and joint decade 2000–2010 task force on neck pain and its associated disorders. Spine 2008;33(4 Suppl.):S39–51.
Hoving JL, Gross AR, Gasner D, Kay T, Kennedy C, Hondras MA, et al. A critical appraisal of review articles on the effectiveness of conservative treatment of neck pain. Spine 2001;26(2):196–205.
Howe DH, Newcombe RG, Wade MT. Manipulation of the cervical spine – a pilot study. Journal of the Royal College of General Practitioners 1983;33:574–9.
Hurwitz EL, Morgenstern H, Harber P, Kominski GF, Yu F, Adams AH. A randomized trial of chiropractic manipulation and mobilization for patients with neck pain: clinical outcomes from the UCLA neck-pain study. Research and Practice 2002;92(10):1634–41.
Hurwitz EL, Carragee JEJ, van der Velde G, Carroll LJ, Nordin M, Guzman J, et al. Treatment of neck pain: noninvasive interventions: results of the bone and joint decade 2000–2010 task force on neck pain and its associated disorders. Spine 2008;33(4 Suppl.):S123–52.
Kanlayanaphotporn R, Chiradejnant A, Vachalathiti R. The immediate effects of mobilization technique on pain and range of motion in patients presenting with unilateral neck pain; a randomized controlled trial. Archives of Physical Medicine and Rehabilitation 2009;90:187–92.
Kanlayanaphotporn R, Chiradejnant A, Vachalathiti R. Immediate effects of the central posteroanterior mobilization technique on pain and range of motion in patients with mechanical neck pain. Disability and Rehabilitation 2010;32(8):622–8.
Kendal MG, Stuart A. The advanced theory of statistics. 2nd ed., vol. 1. New York: Hofner Publishing Co; 1963.
Karlberg M, Magnusson M, Eva-Maj M, Melander A, Moritz U. Postural and symptomatic improvement after physiotherapy in patients with dizziness of suspected cervical origin. Archives of Physical Medicine and Rehabilitation 1996;77:874–82.
Krauss J, Creighton D, Ely JD, Podlewska-Ely J. The immediate effects of upper thoracic translatoric spinal manipulation on cervical pain and range of motion: a randomized clinical trial. The Journal of Manual and Manipulative Therapy 2008;16(2):93–9.
Lee CE, Simmonds MJ, Novy DM, Jones S. Self-reports and clinician-measured physical function among patients with low back pain: a comparison. Archives of Physical Medicine and Rehabilitation 2001;82:227–31.
Linton SJ, Hellsing AL, Hallden K. A population-based study of spinal pain among 35–45 year old individuals. Prevalence, sick leave and health care use. Spine 1998;23(13):1457–63.
MacDermid JC, Walton DM, Avery S, Blanchard A, Etruw E, McAlpine C, et al. Measurement properties of the neck disability index: a systematic review. Journal of Orthopaedic and Sports Physical Therapy 2009;39(5):400–17.
Martinez-Segura R, Fernandez-de-las-Penas C, Ruiz-Saez M, Lopez-Jimenez C, Rodriguez-Blanco C. Immediate effects on neck pain and active range of motion after a single cervical high-velocity low amplitude manipulation in subjects presenting with mechanical neck pain: a randomized controlled trial. Journal of Manipulative and Physiological Therapeutics 2006;29:511–7.
Miller J, Gross A, D'Sylva J, Burnie SJ, Goldsmith CH, Graham N, et al. Manual therapy and exercise for neck pain: a systematic review. Manual Therapy 2010;15(4):334–54.
Muller R, Giles LGF. Long-term follow-up of a randomized clinical trail assessing the efficacy of medication, acupuncture, and spinal manipulation for chronic mechanical spinal pain syndromes. Journal of Manipulative and Physiological Therapeutics 2005;28:3–11.
Nilsson N. A randomized controlled trial of the effect of spinal manipulation in the treatment of cervicogenic headache. Journal of Manipulative and Physiological Therapeutics 1995;18(7):435–40.
Nilsson N, Christensen HW, Hartvigsen J. Lasting changes in passive range of motion after spinal manipulation: a randomized, blind, controlled trial. Journal of Manipulative and Physiological Therapeutics 1996;19(3):165–8.
Nilsson N, Christensen HW, Hartvigsen J. The effect of spinal manipulation in the treatment of cervicogenic headache. Journal of Manipulative and Physiological Therapeutics 1997;20(5):326–30.
Olesen J. Classification and diagnostic criteria for headache disorders, cranial neuralgias and facial pain. Cephalalgia 1988;8(7):61–2.
Olesen J, Gobel H. ICD-10 guide for headaches. Guide to the classification, diagnosis and assessment of headaches in accordance with the tenth revision of the international classification of diseases and related health problems and its application to neurology. Cephalalgia 1997;17(Suppl. 19):29–30.
Parkin-Smith GF, Penter CS. A clinical trial investigating the effect of two manipulative approaches in the treatment of mechanical neck pain: a pilot study. Journal of the Neuromusculoskeletal System 1998;6(1):6–16.
Peeters GGM, Verhagen AP, deBie RA, Oostendorp RAB. The efficacy of conservative treatment in patients with whiplash injury. Spine 2001;26(4):E64–73.
Rubinstein SM, Pool JJM, van Tulder MW, Riphagen II, de Vet HCW. A systematic review of the diagnostic accuracy of provocative tests of the neck for diagnosing cervical radiculopathy. European Spine Journal 2007;16:307–19.
Savolainen A, Ahlberg J, Nummila H, Nissinen M. Active or passive treatment for neck–shoulder pain in occupational health care? A randomized controlled trial. Occupational Medicine 2004;54:422–4.
Schultz KF, Altman DG, Moher D, for the Consort Group. CONSORT 2010 statement: updated guidelines for reporting parallel group randomized trials. Annals Internal Medicine 2010;152(11):1–8, www.consort-statement.org [accessed 7.04.10].
Schumacher HR, Klippel JH, Koopman WJ, editors. Primer on the rheumatic diseases. 10th ed. Atlanta: Arthritis Foundation; 1993.
Sjaastad O, Fredriksen TA, Pfaffenrath V. Cervicogenic headache: diagnostic criteria. Headache 1990;30:725–6.
Skargren EI, Oberg BE. Predictive factors for 1-year outcome of low-back and neck pain in patients treated in primary care: comparison between the treatment strategies chiropractic and physiotherapy. Pain 1998;77:201–7.
Sloop PR, Smith DS, Goldenberg E, Dore C. Manipulation for chronic neck pain: a double-blind controlled study. Spine 1982;7(6):532–5.
Souvlis T, Vicenzino B, Wright A. Neurophysiological effects of spinal manual therapy. In: Boyling JD, Jull GA, editors. Grieve's modern manual therapy, the vertebral column. 3rd ed. Edinburgh: Churchill Livingstone; 2004. p. 367–80.
Spitzer WO, Leblanc FE, Dupuis M. Scientific approach to the assessment and management of activity related spinal disorders. Spine 1987;7:S1–59.
Spitzer WO, Skovron ML, Salmi LR, et al. Scientific monograph of the Quebec task force on whiplash-associated disorders: redefining "whiplash" and its management. Spine 1995;20:S1–73.
Stratford PW, Riddle DL, Binkley JM, Spadoni G, Westaway MD, Padfield B. Using the neck disability index to make decisions concerning individual patients. Physiotherapy Canada 1999 Spring:107–19.

Strunk RG, Hondras MA. A feasibility study assessing manual therapies to different regions of the spine for patients with subacute or chronic neck pain. Journal of Chiropractic Medicine 2008;7:1–8.

van Schalkwyk R, Parkin-Smith GF. A clinical trial investigating the possible effect of the supine cervical rotatory manipulation and the supine lateral break manipulation in the treatment of mechanical neck pain: a pilot study. Journal of Manipulative and Physiological Therapeutics 2000;23 (5):324–31.

Verhagen AP, Scholten-Peeters GGGM, van Wijngaarden S, de Bie R, Bierma-Zeinstra SMA. Conservative treatment for whiplash. Cochrane Database of Systematic Reviews 2007;(2):CD003338. doi:10.1002/14651858.CD003338.pub3 [accessed 20.03.10].

Vernon HT, Aker P, Burns S, Viljakaanen S, Short L. Pressure pain threshold evaluation of the effect of spinal manipulation in the treatment of chronic neck pain: a pilot study. Journal of Manipulative and Physiological Therapeutics 1990;13 (1):13–6.

Vernon HT, Humphreys BK, Hagino CA. A systematic review of conservative treatments for acute neck pain not due to whiplash. Journal of Manipulative and Physiological Therapeutics 2005;28:443–8.

Vernon HT, Humpherys BK, Hagino CA. The outcome of control groups in clinical trials of conservative treatments for chronic mechanical neck pain: a systematic review. BMC Musculoskeletal Disorders 2006;7(58):1–10.

Vernon HT, Humphreys BK, Hagino CA. Chronic mechanical neck disorders in adults treated by manual therapy: a systematic review of change scores in randomized clinical trials. Journal of Manipulative and Physiological Therapeutics 2007;30:215–27.

Wood TG, Colloca CJ, Matthews R. A pilot randomized clinical trial on the relative effect of instrumental (MFMA) versus manual (HVLA) manipulation in the treatment of cervical spine dysfunction. Journal of Manipulative and Physiological Therapeutics 2001;24(4):260–71.

Yurkiw D, Mior S. Comparison of two chiropractic techniques on pain and lateral flexion in neck pain patients: a pilot study. Chiropractic Technique 1996;8(4):155–62.